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From AI Capabilities to Sustainable Competitive Advantages: A Dynamic Capabilities Perspective in E-Commerce and Platform-Based Digital Business

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Abstract

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Contents

List of Abbreviations **v**

List of Figures **vi**

List of Tables **vii**

1 Introduction **1**
1.1 Background of the Study . . . 1
1.2 Research Problem and Research Gap . . . 3
1.3 Research Aim and Questions . . . 4
1.4 Research Propositions . . . 5
1.5 Significance of the Study . . . 6
1.6 Structure of the Thesis . . . 7

2 Literature Review and Theoretical Framework **8**
2.1 AI Capabilities in Digital Business . . . 8
2.2 Resource-Based View and Dynamic Capabilities Theory . . . 10
2.3 Sensing, Seizing and Transforming Capabilities . . . 13
2.4 AI Capabilities and Competitive Advantage . . . 15
2.5 Sustainable Competitive Advantage in E-Commerce and Platform Businesses . . . 18
2.6 Research Gap and Conceptual Framework . . . 20

3 Research Methodology **23**
3.1 Research Philosophy and Design . . . 23
3.2 Literature Search Strategy . . . 24
3.3 Inclusion and Exclusion Criteria . . . 26
3.4 Study Selection and Data Extraction . . . 27
3.5 Quality Assessment and Evidence Synthesis . . . 30
3.6 Methodological Limitations . . . 32

4 Results and Discussion **34**
4.1 Literature Search and Study Characteristics . . . 34
4.2 AI Capabilities and Competitive Advantage . . . 37
4.3 AI Capabilities and Dynamic Capabilities . . . 42
4.4 The Mediating Role of Dynamic Capabilities . . . 45
4.5 Evaluation of Research Propositions . . . 48
4.6 Discussion and Implications of the Findings . . . 50

5 Conclusion and Recommendations **55**
5.1 Summary and Answers to Research Questions . . . 55
5.2 Theoretical Contributions . . . 57

5.3 Managerial Recommendations 58
5.4 Limitations and Future Research 61
5.5 Final Conclusion 63

Bibliography **viii**

A Search Strings **xx**

B Overview of the Included Studies **xxi**

List of Abbreviations

AI	Artificial Intelligence
B2B	Business-to-Business
BDAC	Big Data Analytics Capability
CRM	Customer Relationship Management
DOI	Digital Object Identifier
ESG	Environmental, Social and Governance
GenAI	Generative Artificial Intelligence
MNE	Multinational Enterprise
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RBV	Resource-Based View
SME	Small and Medium-sized Enterprise
VHB	German Academic Association for Business Research (Verband der Hochschullehrer für Betriebswirtschaft)
VRIN	Valuable, Rare, Inimitable, Non-substitutable

List of Figures

2.1 Overview of AI Capabilities in Digital Business 9
2.2 Sensing, Seizing and Transforming Framework 14
2.3 Conceptual Framework of the Study 21

3.1 Literature Screening Flow Diagram 28

4.1 Distribution of Studies by Research Theme 37
4.2 Final Evidence-Informed Conceptual Model 51

List of Tables

2.1 Comparison of the Resource-Based View and Dynamic Capabilities Theory 13
2.2 Summary of Key Theoretical Constructs 17
2.3 Research Gaps Identified in the Literature 20

3.1 Literature Search Strategy 25
3.2 Inclusion and Exclusion Criteria 27
3.3 Data Extraction Framework 29
3.4 Study Quality Assessment Framework 31

4.1 Publication Characteristics of Included Studies 35
4.2 Evidence Linking AI Capabilities to Competitive Advantage 38
4.3 Evidence on Sensing, Seizing and Transforming 43
4.4 Summary of Mediation Evidence 46
4.5 Research Proposition Evaluation 48

5.1 Summary of Answers to the Research Questions 56
5.2 Managerial Recommendations 59

1 Introduction

Few technologies have attracted as much attention from managers and researchers in recent years as artificial intelligence (AI). Online retailers use AI to personalise product recommendations, forecast demand and set prices, while digital platforms rely on it to match users, curate content and automate customer service. Yet the fact that a firm uses AI does not mean that it gains an edge over its competitors. As AI tools become widely available, the decisive question shifts from whether firms adopt AI to how they turn it into an advantage that rivals cannot easily copy, and whether this advantage lasts.

This thesis addresses this question from a dynamic capabilities perspective. Dynamic capabilities describe a firm's ability to sense new opportunities, seize them and transform its resource base as markets change (Teece, 2007, p. 1319). The thesis examines how AI capabilities, understood as the firm's ability to deploy and combine AI-specific resources, relate to these dynamic capabilities, and how both together contribute to sustainable competitive advantage in e-commerce and platform-based digital business. To answer this question, it systematically reviews the peer-reviewed literature on the topic. This chapter first outlines the background of the study, then defines the research problem and gap, states the research aim, questions and propositions, and finally explains the significance and structure of the thesis.

1.1 Background of the Study

AI has moved from an experimental technology to a strategic resource that shapes how firms create and capture value. In the digital era, AI is no longer used only to automate isolated tasks: a systematic review of its strategic use identifies decision support, customer and employee engagement, automation, and new products and services as four sources of value creation (Borges et al., 2021, p. 1). Nevertheless, a further review of the business value of AI concludes that organisations still struggle to adopt and leverage AI and that a coherent understanding of how AI technologies create value is lacking (Enholm et al., 2022, p. 1709). For management research, AI therefore raises a classic strategic question in a new form: under which conditions does a technology become a source of superior and lasting performance?

This question is especially pressing in e-commerce and platform-based digital business. These business models generate large volumes of customer and transaction

data and compete in digital markets whose dynamics depart from conventional competitive theory; here, the scale of a platform's user and complementor network and its distinct platform identity shape its competitive position (Cennamo, 2021, p. 265). AI adds a further layer to this competitive logic. Gregory et al. (2021, p. 534) argue that platforms can benefit from "data network effects": the more a platform learns from the data of its users, the more valuable it becomes to each of them. AI capabilities thus have the potential to reinforce the scale-based advantages that already characterise digital platforms.

First empirical studies from e-commerce support this potential. In a survey of 394 e-commerce firms, Chen et al. (2022, p. 1) find that AI capability, conceptualised as a higher-order construct grounded in the resource-based view, affects firm performance indirectly through firm creativity, AI management and AI-driven decision making. More recently, Bunea & Corboș (2026, p. 1) show for electronic retailing that digital maturity drives AI implementation, which in turn generates benefits that are linked to customer experience and to differentiation- and cost-based competitive advantage. These findings suggest that AI can contribute to competitive outcomes, but they also indicate that the effect runs through organisational conditions rather than through the technology alone.

At the same time, AI is becoming more widely accessible. Cloud services and generative AI make advanced models available to almost any firm, which challenges the assumption that access to AI is in itself a scarce and valuable resource. Using a resource-based view, Krakowski et al. (2023, p. 1425) demonstrate that AI adoption can make traditional competitive capabilities obsolete while creating new sources of heterogeneity in how humans and machines work together. Similarly, Li et al. (2025, p. 10) show that generative AI capabilities contribute to competitive advantage mainly through service innovation, and that this effect depends on environmental dynamism. Notably, Li et al. (2025, p. 10) report the counterintuitive result that high environmental dynamism weakens some of these mediation effects, which suggests that the competitive value of AI capabilities is contingent on the market environment. Both studies point to the need to look beyond AI as a technology and to examine the capabilities that allow firms to deploy, combine and renew it.

Taken together, the background illustrates a tension that motivates this thesis. On the one hand, AI appears to be a powerful source of value in data-rich digital businesses. On the other hand, the growing availability of AI technologies and the dependence of their effects on complementary organisational factors call into question whether AI alone can be the basis of an advantage that competitors cannot easily imitate. Resolving this tension requires a theoretical lens that explains how firms build, deploy and

continuously renew AI-related capabilities, which is why this thesis adopts a dynamic capabilities perspective.

1.2 Research Problem and Research Gap

Despite high expectations and substantial investments, many firms struggle to turn AI into measurable business value. Alnofeli et al. (2026, p. 1) note that although executives treat AI as a strategic priority, only a fraction of organisations report significant returns, particularly in customer-facing functions. Part of this difficulty lies in the nature of AI itself. Its probabilistic outputs, limited transparency and strong dependence on data create implementation challenges that require specific organisational capabilities (Weber et al., 2023, p. 1549). Based on 25 expert interviews, these authors derive four such capabilities: AI project planning and co-development to cope with the inscrutability of AI, and data management and AI model lifecycle management to cope with its data dependency. Such findings indicate that the value of AI depends less on the technology itself than on the organisational capabilities that surround it. The research problem of this thesis therefore concerns the gap between the adoption of AI and the creation of a competitive advantage that firms can sustain over time.

The resource-based view offers a first explanation for this gap. Mikalef & Gupta (2021, p. 1) identify the AI-specific resources that jointly create an AI capability and show that firms with a stronger AI capability achieve higher organisational creativity and performance. However, a resource-based explanation tends to be static. In markets characterised by rapid technological change, such as e-commerce, a lasting advantage depends on the ability to continuously sense opportunities, seize them and reconfigure the resource base (Teece, 2007, p. 1319). Dynamic capabilities theory therefore appears particularly suitable for explaining how AI-related resources are converted into sustainable competitive advantage.

A growing number of studies applies this perspective to AI. Based on case studies of large Norwegian organisations, Mikalef et al. (2021, p. 80) identify micro-foundations through which AI supports sensing, seizing and transforming in business-to-business (B2B) marketing. Quantitative studies report that organisational resilience and strategic change (Liang & Sun, 2026, p. 1) or strategic resilience (Wang & Zhang, 2026, p. 1) mediate the relationship between AI and firm outcomes. Wu et al. (2026, p. 1) add that identical AI investments can lead either to fleeting efficiency gains or to sustained advantages, depending on how managers interpret AI and which layer of dynamic capabilities they activate.

These studies also show that the question of sustainability is far from settled. Liang & Sun (2026, p. 1) explicitly address whether AI enables firms to balance short-term economic returns with long-term sustainable growth, and they find that this effect varies considerably between firms. In a multi-method study of 357 early-stage ventures, Wang & Zhang (2026, p. 1) find that strategic resilience fully accounts for the relationship between AI innovation capability and entrepreneurial success, implying that AI investments create value only when they are converted into adaptive capacity. For e-commerce and platform businesses, in which technologies, customer expectations and competitors change rapidly, it therefore remains unclear whether and how AI capabilities translate into an advantage that persists over time.

Three gaps follow from this literature. First, the evidence is fragmented across industries: much of it stems from manufacturing, B2B marketing, banking or start-ups, while studies specific to e-commerce and digital platforms remain comparatively rare (Chen et al., 2022, p. 1; Bunea & Corboș, 2026, p. 1). Second, many studies measure firm performance or short-term efficiency rather than a competitive advantage that is sustained over time, although these outcomes are conceptually distinct (Wu et al., 2026, p. 1). Third, existing reviews address the business value of AI (Enholm et al., 2022, p. 1709) or AI-driven business model innovation, a field described as fragmented by varied conceptual lenses (Jorzik et al., 2024, p. 1). To the best of the author's knowledge, no systematic review has yet synthesised how AI capabilities, dynamic capabilities and sustainable competitive advantage are linked in e-commerce and platform-based digital business. This thesis addresses this gap.

1.3 Research Aim and Questions

The aim of this study is to systematically review existing academic literature to examine how artificial intelligence capabilities contribute to sustainable competitive advantage in e-commerce and platform-based digital businesses, with particular attention to the role of dynamic capabilities, including sensing, seizing and transforming.

To achieve this aim, the thesis addresses three research questions:

RQ1: How are artificial intelligence capabilities associated with competitive advantage in e-commerce and platform-based digital businesses?

RQ2: How do dynamic capabilities, particularly sensing, seizing and transforming, relate to the relationship between artificial intelligence capabilities and competitive advantage?

RQ3: What factors enable or constrain the development of sustainable competitive advantage through artificial intelligence capabilities in e-commerce and platform-based digital businesses?

RQ1 examines the direct link between AI capabilities and competitive advantage. RQ2 places dynamic capabilities, the theoretical lens of this thesis, at the centre of the analysis. RQ3 asks under which conditions an AI-based advantage can be sustained or why it may erode. The three questions build on each other: RQ1 establishes whether a relationship exists, RQ2 explains the mechanism behind it, and RQ3 identifies the conditions under which the resulting advantage persists. All three questions are answered by systematically reviewing and synthesising peer-reviewed studies (Chapter 3); no primary data are collected. Throughout the thesis, competitive advantage and sustainable competitive advantage are treated as distinct outcomes: a short-term improvement in performance is not automatically interpreted as a sustainable advantage.

1.4 Research Propositions

Since this thesis is based on a systematic literature review rather than on primary data, it formulates research propositions instead of statistically testable hypotheses. The propositions organise the review and are assessed in Section 4.5 as supported, contradicted or inconclusive in light of the reviewed evidence.

P1: Artificial intelligence capabilities are associated with competitive advantage in digital business contexts, although the strength and nature of this relationship may vary across firms and business environments.

P2: Dynamic capabilities, particularly sensing, seizing and transforming, may help explain how firms convert artificial intelligence capabilities into competitive advantage.

P3: The sustainability of AI-enabled competitive advantage may depend on organisational and contextual factors, including the ability to integrate AI into business processes and to adapt to changing competitive conditions.

P1 builds on resource-based studies that link AI capability to firm performance (Mikalef & Gupta, 2021, p. 1) and on survey evidence from supply chain executives that the assimilation of AI improves firm performance, with organisational and customer agility acting as partial mediators (Fosso Wamba, 2022, p. 1). P2 draws on dynamic capabilities theory (Teece, 2007, p. 1319) and on studies in which capability-based mechanisms, such as service innovation, mediate the effect of generative AI capabilities on competitive advantage (Li et al., 2025, p. 10). P3 reflects the argument that AI implementation requires specific organisational capabilities (Weber et al., 2023, p. 1549) and

the finding that the effect of AI on long-term outcomes differs across firm characteristics such as size, industry and life-cycle stage (Liang & Sun, 2026, p. 1).

The propositions are deliberately formulated with caution, since the reviewed studies differ in context, design and measurement. A proposition is considered supported if the majority of the included studies, and especially those of high methodological quality, report findings consistent with it. It is considered contradicted if the evidence predominantly points in the opposite direction, and inconclusive if the findings are mixed or the number of relevant studies is too small to draw a conclusion. Evidence from e-commerce and platform businesses is weighted separately from supporting evidence from other industries.

1.5 Significance of the Study

From a theoretical perspective, the thesis contributes to research at the intersection of strategic management and information systems. It integrates the resource-based view and dynamic capabilities theory to explain not only whether AI capabilities are associated with competitive advantage, but also through which capabilities and under which conditions this advantage can be sustained. By separating evidence on performance, competitive advantage and sustainable competitive advantage, the review also responds to the conceptual ambiguity noted in recent work (Wu et al., 2026, p. 1). Furthermore, the thesis connects the emerging argument that AI shifts the sources of competitive advantage, because it both substitutes and complements human capabilities (Krakowski et al., 2023, p. 1425), with the platform-specific logic of data network effects. In doing so, it examines whether the dynamic capabilities perspective, which was developed for open economies with rapid innovation, also explains how advantages are built and renewed in AI-intensive digital markets. The resulting evidence-informed conceptual model can serve as a starting point for future empirical studies that test the proposed relationships with primary data.

From a managerial perspective, the findings are relevant for managers of e-commerce firms and digital platforms who decide on AI investments. Evidence from electronic retailing shows that digital maturity and AI implementation are key priorities for firms seeking AI-driven advantage (Bunea & Corboș, 2026, p. 1). A structured synthesis of the literature can help managers identify which complementary capabilities need to be developed so that AI investments do not remain isolated technology projects. This is particularly relevant because, as Alnofeli et al. (2026, p. 1) emphasise, substantial investments in AI-powered systems frequently fail to generate meaningful business value.

From a methodological perspective, the thesis applies a transparent and replicable review procedure. The search strategy, selection criteria, screening steps and quality assessment are documented in detail, so that other researchers can reproduce the review and extend it as new studies are published. Given the rapid growth of research on AI in business, such a structured overview of the current state of evidence is a useful basis for future empirical work.

The scope of the thesis is limited to peer-reviewed academic literature on AI capabilities, dynamic capabilities and competitive advantage. Technical questions of AI system design, legal and ethical regulation, and public-sector applications are not examined. Studies from other industries are included only as supporting evidence and are identified as such in the analysis.

1.6 Structure of the Thesis

The thesis is organised in five chapters. Following this introduction, Chapter 2 reviews the literature on AI capabilities in digital business, introduces the resource-based view and dynamic capabilities theory, and discusses the sensing, seizing and transforming capabilities. It then examines the link between AI capabilities and competitive advantage, the specific conditions of e-commerce and platform businesses, and concludes with the research gap and the conceptual framework of the study.

Chapter 3 explains the research philosophy and design of the systematic literature review. It documents the literature search strategy, the inclusion and exclusion criteria, the procedure for study selection and data extraction, the quality assessment and evidence synthesis, and the methodological limitations.

Chapter 4 presents the characteristics of the included studies and synthesises the evidence on AI capabilities and competitive advantage, on AI capabilities and dynamic capabilities, and on the mediating role of dynamic capabilities. On this basis, it evaluates the research propositions and discusses the findings in relation to the literature reviewed in Chapter 2. Finally, Chapter 5 answers the research questions, outlines the theoretical contributions and managerial recommendations, and discusses limitations and avenues for future research.

2 Literature Review and Theoretical Framework

This chapter establishes the theoretical foundation of the thesis. It first clarifies what is meant by AI capabilities in digital business (Section 2.1) and then introduces the two theories that guide the review: the resource-based view and dynamic capabilities theory (Section 2.2). Section 2.3 examines the sensing, seizing and transforming capabilities in more detail, before Section 2.4 reviews the evidence linking AI capabilities to competitive advantage. Section 2.5 discusses what sustainable competitive advantage means in e-commerce and platform-based business. Finally, Section 2.6 summarises the research gaps and derives the conceptual framework of the study.

2.1 AI Capabilities in Digital Business

AI is not a single technology but a wide-ranging set of technologies that promise organisations several advantages in terms of business value (Enholm et al., 2022, p. 1709). To structure this variety, Davenport et al. (2020, p. 24) propose a multidimensional framework that distinguishes AI applications by their level of intelligence, the type of task they perform and whether they are embedded in a robot. More recently, generative AI (GenAI) has added a new class of applications. Cui et al. (2024, p. 583) define GenAI as AI-enabled technology that analyses and learns from existing data and generates novel, humanlike content. According to Grewal et al. (2025, p. 702), GenAI will influence how firms communicate with customers, create marketing content and develop new products and services, although its adoption involves trade-offs regarding the nature of its inputs and the degree of human augmentation required.

For strategic management, the decisive question is not which AI technologies a firm uses, but whether it possesses the capability to deploy them effectively. Mikalef & Gupta (2021, p. 1) therefore shift the focus from AI as a technology to AI capability, which they conceptualise as the joint outcome of several AI-specific resources and which they show to be related to organisational creativity and performance. Case evidence specifies what this capability consists of: in a study of six leading manufacturers, Sjödin et al. (2021, p. 574) identify three critical AI capabilities, namely data pipeline, algorithm development and AI democratisation, and show that scaling them requires firms to innovate their business models. Building on this work, Sjödin et al. (2023, p. 1) conceptualise the perceptive, predictive and prescriptive capacities of AI, which enhance resource efficiency by automating and augmenting data-driven analysis and decision making.

The distinction between automation and augmentation is central to understanding how AI creates value. Raisch & Krakowski (2021, p. 192) argue that, in the management domain, augmentation cannot be neatly separated from automation, because both uses of AI are interdependent across time and space and create a paradoxical tension. Experimental evidence supports the value of combining human and machine work: Przegalinska et al. (2025, p. 1) show that integrating a GenAI tool can significantly improve organisational task performance in areas such as automation, support, creative tasks and innovation, and they highlight the role of task types and AI-related skills.

AI capabilities build on earlier data-related capabilities. Drawing on the resource-based view, Fosso Wamba et al. (2017, p. 356) propose a hierarchical big data analytics capability (BDAC) model and find that it affects firm performance both directly and indirectly through process-oriented dynamic capabilities. In practice, however, firms often lack guidance on how to move from pilot projects to firm-wide use. Haefner et al. (2023, p. 1) note that there is a lack of frameworks for implementing and scaling AI successfully and propose phases through which firms can implement and scale AI in their organisations.

Figure 2.1 integrates these perspectives into an overview of AI capabilities in digital business. It distinguishes the foundations of an AI capability, the capacities that AI provides and the sources of value creation identified by Borges et al. (2021, p. 1): decision support, customer and employee engagement, automation, and new products and services.

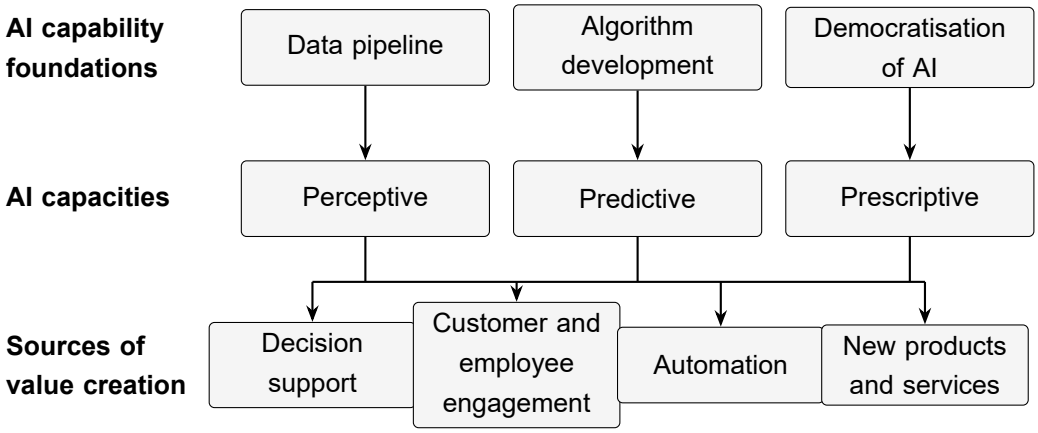


FIGURE 2.1: Overview of AI Capabilities in Digital Business

Source: Author's own work based on Sjödin et al. (2021, p. 574), Sjödin et al. (2023, p. 1) and Borges et al. (2021, p. 1)

In e-commerce and on digital platforms, these capacities materialise in customer-facing applications. Gao et al. (2025a, p. 1) examine how the problem-solving capability of AI chatbots on e-commerce platforms relates to users' confirmation, satisfaction, trust and intention to continue using the service. Yuan et al. (2025, p. 1) examine AI-based

streamers in livestreaming e-commerce and analyse whether AI and human streamers differ in the monetary and non-monetary engagement they generate and how these differences evolve over time. In B2B e-commerce, Shabankareh et al. (2025, p. 1483) find that the service performance and the information and system quality of AI on e-commerce platforms strengthen business users' brand identity, online brand engagement and electronic brand loyalty.

The concept of AI capability is also evolving with the technology itself. Generative AI extends the scope of AI from analysing data to creating content, and it is expected to change how firms communicate with customers, produce marketing content and develop new products and services (Grewal et al., 2025, p. 702). Generative AI is increasingly accessible even to SMEs in emerging economies, which raises the question of how such widely available tools can be linked to firm-specific outcomes (Wishah et al., 2026, p. 1). E-commerce companies in particular are adopting generative AI with the expectation that it will drive revenue growth (Sivathanu & Pillai, 2026, p. 93). The most recent conceptual work already distinguishes generative, predictive and agentic AI as different inputs to AI capability and points to an adoption-value paradox: despite significant investments, many organisations report a lack of business value from AI (Progga & Ali, 2026, p. 1). These developments suggest that AI capability should be understood as a moving target, whose components change as new forms of AI emerge, rather than as a fixed set of resources.

Taken together, the literature conceptualises AI capability at two levels. At the organisational level, it is understood as a bundle of data, technology and human resources that firms must combine and scale (Mikalef & Gupta, 2021, p. 1; Sjödin et al., 2021, p. 574). At the application level, it appears in concrete customer-facing tools whose value depends on how customers perceive and use them (Gao et al., 2025a, p. 1; Shabankareh et al., 2025, p. 1483). This distinction matters for the present thesis. A competitive advantage is unlikely to stem from an individual application, which competitors can often purchase or replicate, but rather from the organisational capability to develop, integrate and continuously improve such applications. It is precisely this organisational level that the resource-based view and dynamic capabilities theory, introduced in the next section, are designed to explain.

2.2 Resource-Based View and Dynamic Capabilities Theory

The resource-based view (RBV) explains differences in firm performance by differences in the resources that firms control. Wernerfelt (1984, p. 171) was among the first to suggest analysing firms from the resource side rather than from the product side and introduced the concept of resource position barriers. Barney (1991, p. 99) formalised

this perspective on the assumptions that strategic resources are heterogeneously distributed across firms and that these differences are stable over time. He identifies four indicators of a resource's potential to generate sustained competitive advantage: value, rareness, imitability and substitutability, commonly summarised as the VRIN criteria. Peteraf (1993, p. 179) integrates these ideas into a model in which four conditions must be met simultaneously for sustained competitive advantage: superior resources (heterogeneity within an industry), ex post limits to competition, imperfect resource mobility and ex ante limits to competition.

The RBV has been widely applied to AI. Besides the AI capability model discussed above (Mikalef & Gupta, 2021, p. 1), Chen et al. (2022, p. 1) construct a higher-order AI capability model for e-commerce firms based on the RBV. Chatterjee et al. (2021, p. 205) combine institutional theory and the RBV to examine how AI-based customer relationship management (CRM) influences firm performance in B2B relationship management, taking firm size, firm age and industry type into account. Yet the RBV logic is also challenged by AI. When AI tools become broadly accessible, their rareness and inimitability become questionable. Cui et al. (2024, p. 583) describe current GenAI systems as democratised technologies that offer only demand-driven insights, and they argue that firms must embed alternative perceptual systems into GenAI platforms to obtain insights that competitors find difficult to replicate. Similarly, Krakowski et al. (2023, p. 1425) show that AI adoption can render traditional capabilities obsolete while new sources of heterogeneity emerge at the intersection of humans and machines.

A further limitation of the RBV is its largely static character. It explains why certain resource positions are valuable, but less how firms build and renew them when markets change. Dynamic capabilities theory addresses this limitation. Teece et al. (1997, p. 509) developed the dynamic capabilities framework to analyse how firms create and capture wealth in environments of rapid technological change. In their view, competitive advantage rests on distinctive organisational processes, shaped by the firm's asset positions and by the evolutionary paths it has adopted or inherited. Whether this advantage erodes depends on the stability of market demand and on the ease of replication and imitation. Teece (2007, p. 1319) later specified the microfoundations of dynamic capabilities, that is, the distinct skills, processes, procedures, organisational structures, decision rules and disciplines that underpin a firm's capacities to sense, seize and reconfigure.

The concept has, however, been debated. Eisenhardt & Martin (2000, p. 1105) argue that dynamic capabilities are specific and identifiable processes, such as product development, strategic decision making and alliancing. Because these processes show significant commonalities across firms, often referred to as best practice, they are more

homogeneous, fungible, equifinal and substitutable than is usually assumed. Moreover, their nature depends on market dynamism: in moderately dynamic markets they resemble stable routines, whereas in high-velocity markets they become simple, experiential and less predictable processes. Two decades later, Schilke et al. (2018, p. 390) note that critics have repeatedly bemoaned the lack of empirical knowledge and the underspecification of the construct, but that research has advanced considerably; they develop a meta-framework that specifies the antecedents, dimensions, mechanisms, moderators and outcomes of dynamic capabilities.

For digital business, the link between dynamic capabilities and business models is particularly relevant. Teece (2018, p. 40) argues that business models, dynamic capabilities and strategy are interdependent: strong dynamic capabilities shape a firm's proficiency in business model design, while the business model in turn influences the firm's dynamic capabilities and constrains the feasibility of particular strategies.

Two recent contributions bridge the resource-based and the capability-based perspective specifically for AI. Building on resource orchestration theory, a multiple case study of Swedish manufacturing SMEs shows that firms structure a portfolio of AI resources by acquiring and accumulating them and then bundle these resources into learning and governance capabilities (Peretz-Andersson et al., 2024, p. 1). At the conceptual level, Kemp (2024, p. 618) introduces the concept of situated AI and argues that firms establish AI-driven competitive advantages through the organisational activities of grounding, bounding and recasting AI, which enable AI-driven capabilities that are firm-specific, cost-effective and appropriate for the opportunities in the strategic environment. Both contributions imply that the competitive relevance of AI does not result from the possession of AI resources, but from how firms orchestrate and embed them, which is precisely the domain of dynamic capabilities.

For the analysis of AI, the two perspectives are best understood as complementary rather than competing. The RBV provides the criteria for assessing whether an AI-related resource can, in principle, be a source of sustained advantage, while dynamic capabilities theory explains how firms build, combine and renew such resources over time. Empirical work on analytics and AI frequently combines both lenses: Fosso Wamba et al. (2017, p. 356) and Mikalef et al. (2020, p. 1) draw on both the RBV and the dynamic capabilities view to explain how analytics capabilities translate into performance. A related development is the resource orchestration perspective, which Zeng et al. (2021, p. 451) describe as a dynamic perspective on the RBV and apply to platform firms whose value creation shifts beyond the boundaries of the firm. The dual lens adopted in this thesis follows this approach: AI capabilities are analysed as

resources in the RBV sense, while their conversion into sustainable competitive advantage is explained through dynamic capabilities. Table 2.1 compares the two theoretical perspectives and indicates what each implies for the study of AI capabilities.

TABLE 2.1: Comparison of the Resource-Based View and Dynamic Capabilities Theory

Dimension	**Resource-Based View**	**Dynamic Capabilities Theory**
Key sources	Wernerfelt (1984, p. 171); Barney (1991, p. 99); Peteraf (1993, p. 179)	Teece et al. (1997, p. 509); Eisenhardt & Martin (2000, p. 1105); Teece (2007, p. 1319)
Core assumption	Resources are heterogeneously distributed and imperfectly mobile	Firms must continuously adapt their resource base to rapid technological and market change
Source of advantage	Resources that are valuable, rare, inimitable and non-substitutable	Distinctive processes, asset positions and paths; sensing, seizing and reconfiguring
View of time	Largely static: explains advantageous resource positions	Dynamic: explains how resource positions are built, renewed and protected
Main criticism	Limited explanation of change in dynamic markets	Construct underspecified; capabilities may be similar across firms (best practice)
Implication for AI	AI resources may lose value as AI tools become widely available	Advantage depends on how AI is used to sense, seize and transform

Source: Author's own work based on Barney (1991, p. 99), Peteraf (1993, p. 179), Teece et al. (1997, p. 509), Eisenhardt & Martin (2000, p. 1105) and Schilke et al. (2018, p. 390)

2.3 Sensing, Seizing and Transforming Capabilities

Following Teece (2007, p. 1319), dynamic capabilities comprise three clusters of activities. Sensing refers to identifying and assessing opportunities and threats in the environment; seizing refers to mobilising resources to address these opportunities and to capture value from them; and transforming, also called reconfiguring, refers to the continuous renewal of the firm's tangible and intangible assets and organisational structures. Together, these capacities allow firms to create, deploy and protect the intangible assets that support superior long-run performance. Figure 2.2 illustrates this framework and the role that AI can play in each of the three activities.

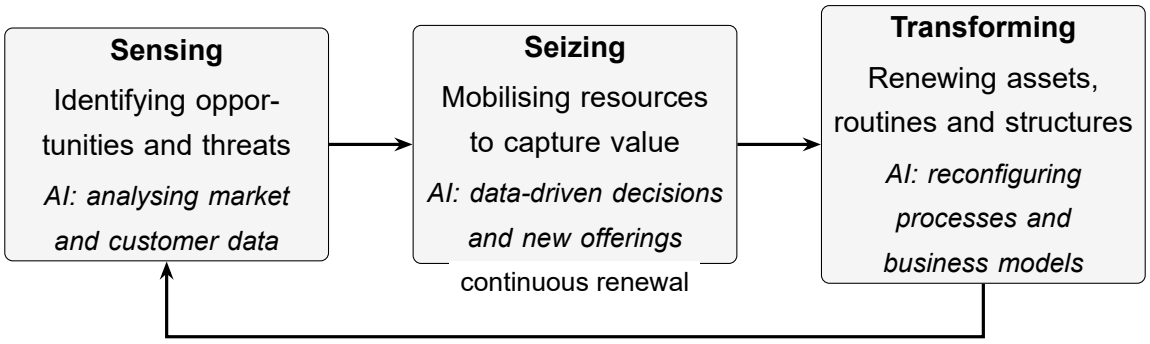


FIGURE 2.2: Sensing, Seizing and Transforming Framework

Source: Author's own work based on Teece (2007, p. 1319) and Hossain et al. (2022, p. 240)

Research on digital transformation has applied this framework to technological change more broadly. In a qualitative study of incumbent firms, Warner & Wäger (2019, p. 326) propose a process model of nine microfoundations that trigger, enable and hinder the building of dynamic capabilities for digital transformation, which they describe as an ongoing process of strategic renewal with agility as its core mechanism. Vial (2019, p. 118), reviewing 282 works, conceptualises digital transformation as a process in which digital technologies create disruptions that trigger strategic responses from organisations, and he explicitly calls for research on the role of dynamic capabilities. Volberda et al. (2021, p. 1) add that digital transformation efforts fail without the right mindset, appropriate digital routines and structural changes, highlighting the interplay between managerial cognitive barriers, the reconfiguration of routines and new organisational forms.

AI-specific studies show how each dynamic capability can be supported by analytics and AI. Hossain et al. (2022, p. 240) find that a firm's marketing analytics capability plays a vital role in sensing, seizing and reconfiguring the market, leading to a sustained competitive advantage, and that the adoption of AI on the basis of the marketing analytics platform further increases these capabilities. Similarly, Mikalef et al. (2021, p. 80) identify AI-specific micro-foundations of dynamic capabilities that help organisations manage B2B marketing in dynamic and uncertain environments.

The three capabilities do not necessarily contribute in the same way. Using panel data on Chinese listed firms, Huang & Yang (2026, p. 1) show that sensing and seizing capabilities mainly strengthen efficiency-oriented, data-driven value propositions, whereas reconfiguring capability is particularly important for product-oriented value propositions. They describe the tendency of AI knowledge to favour efficiency gains as an efficiency trap that firms can only overcome through reconfiguration. This finding is relevant for the question of sustainability, since efficiency gains are typically easier to imitate than new products and business models.

Studies on big data analytics provide additional support for the mediating role of dynamic capabilities. Mikalef et al. (2019, p. 272) propose that BDAC enables firms to generate insights that strengthen their dynamic capabilities, which in turn positively affect incremental and radical innovation capabilities. In a subsequent study of 202 Norwegian chief information officers and IT managers, Mikalef et al. (2020, p. 1) find that BDAC is related to competitive performance indirectly, through dynamic capabilities that strengthen marketing and technological capabilities. Finally, Sjödin et al. (2023, p. 1) identify novel dynamic capabilities that underpin the innovation of AI-enabled business models, indicating that AI does not only rely on dynamic capabilities but also requires new ones.

Two critical observations follow from this stream of research. First, the relationship between AI and dynamic capabilities appears to be reciprocal: AI can strengthen sensing, seizing and transforming, but exploiting AI at scale itself requires new dynamic capabilities, such as the capability to innovate AI-enabled business models (Sjödin et al., 2023, p. 1). Second, the microfoundations of dynamic capabilities include managerial cognition. Without the appropriate mindset, digital initiatives tend to fail (Volberda et al., 2021, p. 1), and the way in which managers frame AI determines which layer of dynamic capabilities a firm activates (Wu et al., 2026, p. 1). Consequently, dynamic capabilities cannot be reduced to technical or procedural routines. This also reflects the measurement challenge noted by Schilke et al. (2018, p. 390): studies operationalise dynamic capabilities in very different ways, which limits the comparability of their findings and must be considered when the evidence is synthesised in Chapter 4.

2.4 AI Capabilities and Competitive Advantage

A growing body of empirical research links AI capabilities to firm-level outcomes. Besides the positive relationship between AI capability and performance reported by Mikalef & Gupta (2021, p. 1), Fosso Wamba (2022, p. 1) shows, based on a survey of supply chain executives, that AI assimilation predicts firm performance and that organisational and customer agility act as complementary partial mediators. In customer-facing functions, Alnofeli et al. (2026, p. 1) conceptualise AI-powered CRM as a higher-order organisational capability and find that it enhances marketing ambidexterity, which in turn improves profitability and competitive advantage. Chatterjee et al. (2021, p. 205) similarly examine how AI-based CRM affects firm performance and competitive advantage in the B2B context.

Several studies emphasise that this relationship is mediated rather than direct. Li et al. (2025, p. 10) find that service concept, service process and customer experience

innovation mediate the relationship between GenAI capabilities and competitive advantage, and that environmental dynamism weakens some of these mediation effects. Dinh (2026, p. 84) position AI-enabled knowledge capabilities as dynamic capabilities and show that they enhance new product performance both directly and indirectly through digital product innovation radicalness, with digital sustainability leadership acting as a boundary condition. Strategic resilience (Wang & Zhang, 2026, p. 1) as well as organisational resilience and strategic change (Liang & Sun, 2026, p. 1) have also been identified as mechanisms through which AI translates into favourable outcomes.

Beyond these mediating mechanisms, AI may change the very sources of competitive advantage. Krakowski et al. (2023, p. 1425) show that neither humans nor AI alone explain performance differences once AI is adopted; instead, a new decision-making resource emerges at the human–AI intersection. Wu et al. (2026, p. 1) add a cognitive perspective, arguing that managers frame AI as a tool, an actor or a strategic resource, and that these framings determine which layer of dynamic capabilities a firm can activate and whether it achieves fleeting efficiency gains or sustained strategic advantages.

The transformation of business models is a further pathway, but also a demanding one. Black et al. (2024, p. 1) observe that, while emerging organisations use AI as a key basis of competitive advantage, relatively few established organisations have fundamentally transformed their business models through AI; based on case studies with executives of ten organisations, they identify factors linked to successful AI-enabled business model transformation. Jorzik et al. (2024, p. 1), reviewing 180 articles, describe research on AI-driven business model innovation as fragmented due to varied conceptual lenses and units of analysis.

Finally, the benefits of AI depend on contextual conditions. Dai & Zhang (2026, p. 1) show for 2,986 Chinese listed firms that AI adoption improves supply chain resilience through greater transparency, operational efficiency and optimised supply chain concentration, but that these effects depend on institutional support and vary across firm size and ownership structure. Ali et al. (2025, p. 1) find that AI capabilities foster organisational sustainability through green human resource management, with digital literacy strengthening this indirect effect. Together with the implementation capabilities described by Weber et al. (2023, p. 1549), these studies suggest that AI capabilities create value only in combination with complementary organisational conditions.

A critical look at the designs of these studies qualifies their findings. Much of the quantitative evidence relies on cross-sectional surveys of managers analysed with structural equation modelling, for example with samples of 205 supply chain executives (Fosso Wamba, 2022, p. 1), 260 firms (Li et al., 2025, p. 10) or 394 e-commerce firms (Chen et al., 2022, p. 1). Such designs can establish associations but provide limited evidence

of causality, and they depend on managers' self-assessments of both capabilities and outcomes. Panel studies based on archival data from listed firms address some of these concerns, but they are concentrated in China (Liang & Sun, 2026, p. 1; Huang & Yang, 2026, p. 1; Dai & Zhang, 2026, p. 1), which limits their transferability to European e-commerce markets. Qualitative case studies, in turn, provide rich insights into mechanisms but are based on small numbers of firms (Sjödin et al., 2021, p. 574; Black et al., 2024, p. 1). These differences in design, context and measurement have to be taken into account when the evidence is assessed in the systematic review. Table 2.2 summarises the key theoretical constructs used in this thesis.

TABLE 2.2: Summary of Key Theoretical Constructs

Construct	**Understanding in this thesis**	**Key sources**
AI capability	The firm's ability to deploy AI, emerging from the joint combination of AI-specific resources such as data pipelines, algorithm development and AI democratisation	Mikalef & Gupta (2021, p. 1); Sjödin et al. (2021, p. 574)
Dynamic capabilities	Capacities to sense, seize and reconfigure, grounded in skills, processes, structures and decision rules	Teece et al. (1997, p. 509); Teece (2007, p. 1319)
Sensing	Identifying and assessing opportunities and threats, for example through analysing market and customer data	Teece (2007, p. 1319); Hossain et al. (2022, p. 240)
Seizing	Mobilising resources and designing business models to capture value from opportunities	Teece (2007, p. 1319); Teece (2018, p. 40)
Transforming	Continuous renewal of assets, routines and organisational structures	Teece (2007, p. 1319); Huang & Yang (2026, p. 1)
Competitive advantage	Superior performance relative to competitors, e.g. through differentiation or cost leadership	Barney (1991, p. 99); Bunea & Corboș (2026, p. 1)
Sustainable competitive advantage	An advantage that persists because it cannot easily be imitated or substituted and is continuously renewed	Barney (1991, p. 99); Peteraf (1993, p. 179); Teece (2007, p. 1319)
Data network effects	A platform becomes more valuable to each user the more it learns from user data	Gregory et al. (2021, p. 534); Chen et al. (2026, p. 187)

Source: Author's own work

2.5 Sustainable Competitive Advantage in E-Commerce and Platform Businesses

E-commerce firms and digital platforms compete under specific conditions. In digital markets, the competitive position of a platform depends on its size, reflecting the scale of its user and complementor network, and on its distinct platform identity (Cennamo, 2021, p. 265). AI strengthens a further mechanism: through data network effects, a platform becomes more valuable to each user the more it learns from the data it collects (Gregory et al., 2021, p. 534). These mechanisms can create self-reinforcing advantages that are difficult for smaller competitors to match.

However, recent theory suggests that network effects alone do not secure a lasting advantage. Chen et al. (2026, p. 187) argue that the value of network effects depends on matching quality, which is a function of platform-specific algorithm technology and market-level data-driven learning. Their model shows that an entrant with a superior algorithm may outcompete an incumbent with a larger user base, a strategy they call algorithm envelopment. For platform businesses, this implies that a sustainable advantage depends on the continuous development of AI capabilities rather than on an initial user base.

The resource-based logic also has to be extended beyond the firm's boundaries. Studying sharing economy platforms, Zeng et al. (2021, p. 451) identify three mechanisms through which platform firms orchestrate external resources, that is, crowds of suppliers and consumers, to gain a competitive advantage: constructing on-demand resource adaptation, building big-data-driven network effects and enabling ecosystem resource coordination. For B2B platforms, Hautamäki & Heikinheimo (2025, p. 1) show that platform owners must continuously renew their capabilities and shift their mindset throughout the platform lifecycle, and that they increasingly use AI tools to innovate platform-based business models.

The orchestration perspective has recently been extended to international business. Teece & Pitelis (2026, p. 689) argue that the emergence of digital platforms, business ecosystems and AI challenges internalisation theory, whose explanatory power is increasingly constrained in environments characterised by ecosystem-based value creation and technologically mediated forms of control. They therefore propose orchestration theory, according to which modern multinational enterprises are best understood as focal firms that orchestrate globally distributed resources and partners rather than simply internalising transactions. For smaller firms, the relationship is reversed: they participate in platforms orchestrated by others. A study of Chinese SMEs using cross-border e-commerce platforms examines how the coupling of AI adoption and strategic

agility shapes their position in global value chains and shows that the depth and breadth of platform embeddedness condition this relationship (Guo et al., 2025, p. 1250). In platform-based business, competitive advantage therefore depends not only on a firm's own AI capabilities but also on its position in, and its ability to orchestrate or leverage, a wider ecosystem.

In e-commerce, first studies confirm that AI can contribute to performance and competitive advantage. Besides the evidence from e-commerce firms (Chen et al., 2022, p. 1) and electronic retailing (Bunea & Corboș, 2026, p. 1), Aljarboa (2024, p. 1) draws on the dynamic capabilities framework and finds that the adoption of AI in e-commerce enhances the business performance of small and medium-sized enterprises (SMEs). With regard to long-term outcomes, Wang & Zhang (2025, p. 1) show for TikTok online retailers that the adoption of AI digital employees promotes exploitative and exploratory green innovation, which in turn enhances future sustainable innovation performance.

The sustainability of an AI-based advantage in e-commerce also depends on customers. Vu et al. (2026, p. 1) find that consumer trust in AI-enabled e-commerce is a continuously adjusted, experience-based judgement shaped by the personalisation–privacy paradox, that is, the tension between desiring tailored recommendations and resisting the data practices needed to produce them. Since AI-based personalisation depends on customer data, a loss of trust can weaken the data basis on which data network effects rest. Customer responses to AI may also change over time, as research on the evolving role of AI streamers in livestreaming e-commerce suggests (Yuan et al., 2025, p. 1), and the continued use of AI chatbots is linked to confirmed expectations, satisfaction and trust (Gao et al., 2025a, p. 1). A sustainable competitive advantage in e-commerce therefore requires firms not only to build AI capabilities, but also to renew them continuously and to maintain the trust of their customers.

Relating these findings back to the RBV criteria clarifies the specific conditions of sustainability in this context. AI technologies as such are increasingly available and therefore rarely rare or inimitable (Cui et al., 2024, p. 583). What may be more difficult to imitate are the firm-specific combinations of proprietary data, algorithm quality and learning effects (Chen et al., 2026, p. 187), the orchestration of external resources in platform ecosystems (Zeng et al., 2021, p. 451) and the trust relationships with customers on which data access depends (Vu et al., 2026, p. 1). At the same time, these advantages are exposed to erosion, since a competitor with a better algorithm may overcome an incumbent's user base. In line with Teece et al. (1997, p. 509), sustainable competitive advantage in e-commerce and platform-based business is therefore best understood not as a stable position, but as the result of the continuous renewal of AI-related resources through dynamic capabilities.

2.6 Research Gap and Conceptual Framework

The review shows that the relationship between AI capabilities and competitive advantage has received considerable attention, but it also reveals several gaps, which are summarised in Table 2.3. Most importantly, the evidence is dominated by studies from manufacturing, B2B marketing, banking and supply chains, while e-commerce and platform contexts remain comparatively underrepresented. Many studies measure firm performance rather than sustainable competitive advantage, and dynamic capabilities are often treated as an aggregate construct, although recent evidence suggests that sensing, seizing and transforming contribute in different ways (Huang & Yang, 2026, p. 1).

TABLE 2.3: Research Gaps Identified in the Literature

No.	Research gap	Indicative sources	Addressed by
G1	Evidence is concentrated in manufacturing, B2B, banking and supply chains; e-commerce and platform contexts are underrepresented	Sjödin et al. (2021, p. 574); Alnofeli et al. (2026, p. 1); Dai & Zhang (2026, p. 1)	RQ1; core vs. supporting studies
G2	Performance, competitive advantage and sustainable competitive advantage are often not distinguished	Wu et al. (2026, p. 1); Liang & Sun (2026, p. 1)	RQ1, RQ3
G3	Dynamic capabilities are often aggregated, although sensing, seizing and transforming may contribute differently	Huang & Yang (2026, p. 1); Hossain et al. (2022, p. 240)	RQ2; P2
G4	Platform-specific and customer-related boundary conditions are rarely integrated	Chen et al. (2026, p. 187); Vu et al. (2026, p. 1)	RQ3; P3
G5	Existing reviews focus on AI business value or business model innovation, not on the AI–dynamic capabilities–sustainable advantage link	Enholm et al. (2022, p. 1709); Jorzik et al. (2024, p. 1)	Systematic review (Chapter 3)

Source: Author's own work

Based on the theories and findings reviewed above, Figure 2.3 presents the conceptual framework of the study. AI capabilities form the starting point. Their direct association with competitive advantage corresponds to proposition P1. Following dynamic capabilities theory (Teece, 2007, p. 1319), AI capabilities are expected to strengthen sensing, seizing and transforming, which in turn enable firms to turn temporary advantages into sustainable ones (P2). Organisational and contextual factors, such as the integration

of AI into business processes (Weber et al., 2023, p. 1549), environmental dynamism (Li et al., 2025, p. 10), platform competition (Chen et al., 2026, p. 187) and customer trust (Vu et al., 2026, p. 1), are expected to influence whether this advantage persists (P3).

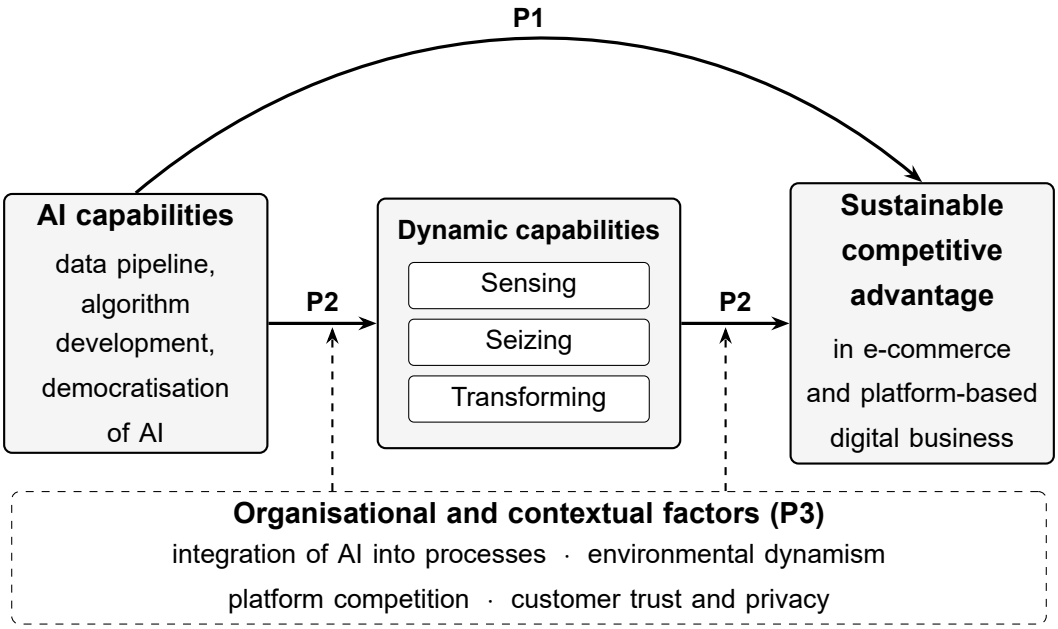


FIGURE 2.3: Conceptual Framework of the Study

Source: Author's own work based on Teece (2007, p. 1319) and Mikalef & Gupta (2021, p. 1)

Three assumptions underlie the framework. First, AI capabilities are treated as an organisational capability rather than as individual technologies or applications, in line with the capability-based view of AI (Mikalef & Gupta, 2021, p. 1). Second, the three dynamic capabilities are modelled separately, since sensing, seizing and transforming may contribute to different types of value creation (Huang & Yang, 2026, p. 1). Third, the framework distinguishes between competitive advantage, which may be temporary, and sustainable competitive advantage, which requires that the advantage resists imitation and is continuously renewed (Barney, 1991, p. 99; Teece, 2007, p. 1319). The direct path from AI capabilities to competitive advantage (P1) is therefore expected to explain short-term gains, whereas the path through dynamic capabilities (P2) is expected to be decisive for sustainability. The contextual factors (P3) capture the conditions that the reviewed literature identifies as enabling or constraining this process.

It should be noted that the framework is not a causal model to be tested statistically. Instead, it serves as an analytical lens that structures the data extraction and the synthesis of the evidence in the systematic literature review. Each included study will be

mapped to the elements of the framework, which allows the propositions to be evaluated as supported, contradicted or inconclusive.

The framework links the research questions to the theory: RQ1 addresses the relationship between AI capabilities and competitive advantage, RQ2 the role of sensing, seizing and transforming, and RQ3 the organisational and contextual factors that enable or constrain the sustainability of the advantage. The following chapter describes the systematic literature review through which the framework is assessed against the available evidence.

3 Research Methodology

This chapter describes how the research questions are answered. It first explains the research philosophy and the choice of a systematic literature review as research design (Section 3.1). It then documents the literature search (Section 3.2), the inclusion and exclusion criteria (Section 3.3), the selection of studies and the extraction of data (Section 3.4), and the procedures for quality assessment and evidence synthesis (Section 3.5). The chapter closes with a discussion of the methodological limitations (Section 3.6).

3.1 Research Philosophy and Design

The thesis follows a pragmatist research philosophy. Its aim is not to test a single causal model, but to understand how AI capabilities, dynamic capabilities and sustainable competitive advantage are linked, based on the best available evidence. This evidence is heterogeneous: it includes quantitative surveys, panel studies of listed firms, qualitative case studies, formal models and conceptual papers. A pragmatist stance allows these different types of knowledge to be combined as long as they contribute to answering the research questions. Ontologically, the thesis assumes that capabilities and competitive advantages are real organisational phenomena, but that they can only be observed indirectly through the concepts, measures and interpretations used in the primary studies. The review therefore treats the findings of each study as evidence that has to be interpreted in light of its theoretical lens, its context and its design.

The research logic is abductive. On the one hand, the review is guided by the theoretical framework developed in Chapter 2 (Figure 2.3) and by the three propositions formulated in Section 1.4. On the other hand, the synthesis remains open to themes that are not anticipated by the framework, for example new mechanisms or boundary conditions. In this way, deductive elements (assessing the propositions) are combined with inductive elements (identifying emerging themes).

As research design, the thesis uses a systematic literature review. Tranfield et al. (2003, p. 207) argue that traditional narrative reviews frequently lack thoroughness and can be biased by the researcher, and they propose systematic review as a methodology for developing evidence-informed management knowledge. Snyder (2019, p. 333) similarly notes that knowledge production in business research is fragmented and interdisciplinary, and that traditional reviews are often conducted ad hoc rather than following a specific methodology. Among the different types of reviews, which Paré et al. (2015,

p. 183) distinguish in a typology of nine review types along seven dimensions, this thesis adopts a systematic review with a qualitative, thematic synthesis. Its procedure follows the guidelines of Kraus et al. (2022, p. 2577) and Paul & Criado (2020, p. 1), who describe the purpose, steps and structure of systematic literature reviews in management and international business research.

The choice of this design has several advantages. First, it fits the research gap identified in Section 2.6: the evidence on AI capabilities and competitive advantage is fragmented across industries, theories and methods, and a systematic synthesis is needed to integrate it. Second, the explicit documentation of every step makes the review transparent and replicable, which reduces the risk of a selective use of sources. Third, a review can draw on a much larger evidence base than a single empirical study could provide within the scope of a Master's thesis. At the same time, the design has drawbacks. A review cannot establish causality beyond what the primary studies allow, it depends on what has been published and indexed, and its conclusions are only as reliable as the underlying studies. Collecting primary data, for example through a survey of e-commerce managers, would have allowed specific measurement of sustainable competitive advantage, but it would have provided evidence from only one sample at one point in time. Given the aim of the thesis, the systematic review is therefore considered the more suitable design. Its limitations are addressed through the quality assessment (Section 3.5) and discussed in Section 3.6.

3.2 Literature Search Strategy

The literature search was carried out on 30 October 2026 in the EBSCO Discovery Service, accessed through the university library of the IU International University of Applied Sciences. The discovery service searches a large number of databases in a single interface; in this search, records were retrieved among others from Business Source Ultimate, ScienceDirect, Emerald Insight, Springer Nature Journals, Academic Search Ultimate and the Complementary Index. A discovery service was chosen because it covers the main publishers of management and information systems journals while allowing a single, consistent search syntax.

The search terms were derived from the research questions and organised into concept blocks. The first block captures AI capabilities, the second block captures the theoretical outcomes of interest (dynamic capabilities and competitive advantage), and the third block captures the business context of the thesis. Synonyms within a block were combined with `OR`, and the blocks were combined with `AND`. Truncation (`*`) was used to capture variants such as *capability* and *capabilities*. To keep the results focused, all terms were searched in the title (`TI`) and abstract (`AB`) fields only. Two filters were

applied: peer-reviewed publications and a publication period from 2015 to 2026. The starting year reflects the fact that research on AI as an organisational capability only gained momentum in the second half of the 2010s.

Two searches were conducted, as summarised in Table 3.1; the complete search strings are documented in Appendix A. The primary search (S1) combined AI capability terms with dynamic capabilities and competitive advantage, irrespective of industry, and returned 407 records. Because studies set in e-commerce and platform businesses do not always use the term *AI capability*, a supplementary search (S2) combined broader AI terms with the outcome block and the e-commerce and platform context block; it returned 205 records. Testing an even broader string without the capability or context restriction yielded 4,368 peer-reviewed records, which would not have been manageable with the screening resources of a single reviewer and would have contained a large share of irrelevant studies. As Hiebl (2023, p. 229) emphasises, the sample selection process is a vital part of systematic reviews in management research and must be organised and reported transparently; the two-step search strategy was chosen for this reason.

TABLE 3.1: Literature Search Strategy

Search	**Concept blocks (searched in title and abstract)**	**Records**	**After automatic de-duplication**
S1 (primary)	AI capability terms **AND** dynamic capabilities / (sustainable) competitive advantage	407	214
S2 (supplementary)	Broad AI terms (incl. generative AI) **AND** dynamic capabilities / competitive advantage **AND** e-commerce / online retail / digital platform terms	205	106
Filters	Peer reviewed; publication years 2015–2026; search date 30 October 2026		
Database	EBSCO Discovery Service via the IU university library	**612**	**320**

Source: Author's own work; full search strings in Appendix A

In addition to the database search, 16 studies were identified through other methods. These are studies that had already been located in the library during the literature review in Chapters 1 and 2 and that meet the inclusion criteria, but were not retrieved by the search strings, for example because their abstracts use terms such as *AI assimilation* or *AI-based CRM* instead of *AI capability*. Their inclusion follows the advice of Webster & Watson (2002, p. xvi) not to rely on keyword searches alone but to also draw on the

citations of key articles. In line with the PRISMA 2020 statement (Page et al., 2021, p. 178), these studies are reported separately in the flow diagram.

3.3 Inclusion and Exclusion Criteria

The eligibility of each record was assessed against predefined criteria, which are summarised in Table 3.2. Each exclusion was assigned a reason code (E1 to E7 and E9), so that the selection process can be traced in the screening log. Four criteria require explanation.

First, the review is restricted to firm-level studies of business organisations, since the research questions concern the competitive advantage of firms. Studies on public organisations, individual employees, consumers or national economies were excluded (E3). Second, the term *sustainable* is used in this thesis in the strategic sense of an advantage that persists over time. Studies whose outcome is limited to environmental, green or ESG performance, without any reference to competitive advantage or firm performance, were therefore excluded (E7), even though they may use the word *sustainable*. Third, studies on AI were only included if AI capabilities or the organisational use of AI were a focal construct and were related to dynamic capabilities, competitive advantage or firm performance (E4).

Fourth, the same eligibility criteria were applied to all studies, irrespective of the journal in which they were published. Journal rankings were deliberately not used as an inclusion or exclusion criterion, because a study should neither be included merely because it appears in a highly ranked journal nor excluded merely because its journal is not ranked. Instead, the listing of each journal in the VHB Rating 2024 of the German Academic Association for Business Research is recorded as descriptive information, and the quality of each study is assessed on the basis of its methodology and evidence (Section 3.5). The only exception concerns business articles published in journals far outside their stated subject scope, such as business studies in a physiotherapy journal. Such cases indicate doubtful peer review and were excluded (E9).

Finally, the included studies are classified into two tiers. Core studies examine e-commerce, online retail or platform-based businesses, while supporting studies examine AI capabilities in other industries. This distinction does not affect inclusion, but it ensures that evidence from other industries is not mistaken for context-specific evidence in the synthesis.

TABLE 3.2: Inclusion and Exclusion Criteria

Criterion	Inclusion	Exclusion (code)
Publication type	Peer-reviewed journal articles	Conference papers, book chapters (E1); editorials, introductions, practitioner or descriptive pieces (E5); literature reviews and bibliometric studies (E6)
Language	English	Other languages (E2)
Period	Published 2015–2026	Earlier publications (search filter)
Level of analysis	Business firms, including SMEs, start-ups, MNEs and platform firms	Public sector, individuals, consumers, national or macro level (E3)
Focal construct	AI capabilities or organisational AI use as a focal construct	AI not focal or not related to the research questions (E4)
Outcome or mechanism	Dynamic capabilities, competitive advantage or firm performance	Only environmental, green or ESG outcomes (E7)
Outlet	Journal within its stated subject scope; VHB listing recorded descriptively only	Business article in a journal far outside its subject scope (E9)
Context (tier)	Core: e-commerce, online retail and platforms; supporting: other industries	– (classification only)

Source: Author's own work

3.4 Study Selection and Data Extraction

The selection of studies followed the PRISMA 2020 statement, which provides a reporting guideline for the identification, screening and inclusion of studies in systematic reviews (Page et al., 2021, p. 178). The process is shown in Figure 3.1. Of the 612 records identified, 292 duplicates were removed by the automatic de-duplication of the EBSCO Discovery Service, which merges identical records from different databases. A further 44 duplicates, which appeared in both searches or had not been merged automatically, were removed manually by comparing DOIs and titles. The remaining 276 records were screened on the basis of their titles and abstracts, and 140 records were excluded. The most frequent reasons were that AI capabilities were not the focal construct or were unrelated to the research questions (49), that the records were conference papers (28), that the outcome was limited to environmental performance (23) and that the articles were not written in English (18).

The remaining 136 reports were assessed for eligibility on the basis of their abstracts, keywords and bibliographic information. In this step, 43 reports were excluded, most

often because AI capabilities were not the focal construct or the study did not address dynamic capabilities, competitive advantage or firm performance (22), because they were essays or practitioner pieces rather than research articles (7), or because their outcome was limited to environmental performance (7). This resulted in 93 studies from the database search. Together with the 16 studies identified through other methods, the review includes 109 studies, of which 19 are core studies from e-commerce and platform contexts and 90 are supporting studies from other industries. Of the 109 studies, 56 are published in journals listed in the VHB Rating 2024. All screening decisions and reasons are documented in a screening log, which is provided as a supplementary spreadsheet.

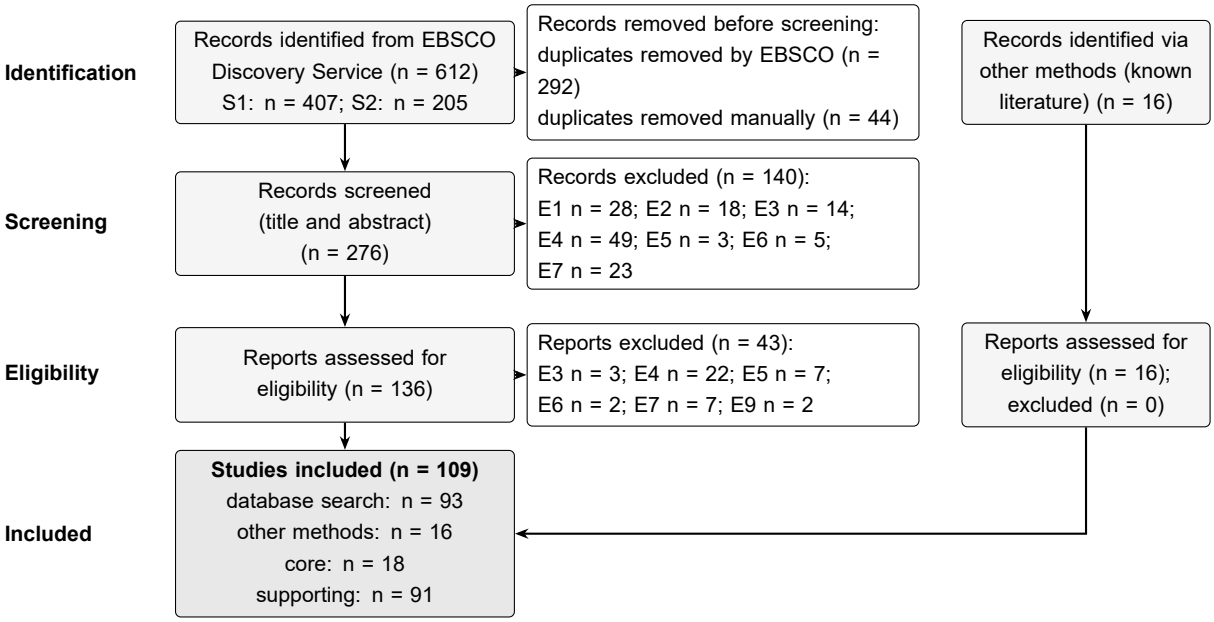


FIGURE 3.1: Literature Screening Flow Diagram

Source: Author's own work based on the PRISMA 2020 flow diagram (Page et al., 2021, p. 178)

For each included study, data are extracted into a structured spreadsheet using the framework shown in Table 3.3. The data items correspond to the elements of the conceptual framework, so that every study can be mapped to the relationships that the propositions address. Besides bibliographic information, the extraction covers the research design and context, the conceptualisation of AI capabilities, the role of sensing, seizing and transforming, the type of outcome and the main findings. To address the conceptual distinction emphasised in Section 1.3, the outcome of each study is classified as firm performance, competitive advantage or sustainable competitive advantage.

TABLE 3.3: Data Extraction Framework

Data item	Description
Bibliographic data	Authors, year, journal, tier (core or supporting), VHB listing
Research objective and theory	Aim of the study; theoretical lens (RBV, dynamic capabilities or other)
Research design	Survey, panel data, case study, experiment, formal model or conceptual paper
Sample and context	Sample size, country, industry and business context
AI capability construct	How AI capability or AI use is conceptualised and measured
Dynamic capabilities	Role of sensing, seizing and transforming (or related constructs such as agility and resilience)
Outcome type	Firm performance, competitive advantage or sustainable competitive advantage
Main findings	Direction and nature of the reported relationships
Mediators and moderators	Mechanisms and boundary conditions reported in the study
Relevance to propositions	Evidence supporting or contradicting P1, P2 and P3
Limitations	Limitations reported by the authors or identified in the quality assessment

Source: Author's own work

The data were extracted from the abstracts, keywords and bibliographic records of the included studies, which were retrieved through the library and through the open bibliographic databases Crossref, OpenAlex and Semantic Scholar. To make the coding comparable across studies, several data items were coded with predefined categories. The outcome of each study was classified as firm performance, competitive advantage, sustainable competitive advantage, resilience, innovation, business model innovation or another outcome. The role of the three dynamic capabilities was recorded separately for sensing, seizing and transforming, so that a study could address one, two or all three of them; related constructs such as agility were assigned to the capability they primarily represent. For each proposition, the evidence of a study was coded as supportive, conditional, not significant, constraining or not addressed. A study was coded as conditional when the authors explicitly described the relationship as depending on specific circumstances, and as not significant when the direct effect of AI capabilities was reported as statistically insignificant, even if indirect effects were present.

Firm performance was not treated as competitive advantage, and competitive advantage was only coded as sustainable if the authors explicitly framed it as sustained or

long-term. Where a study addressed several outcomes, the most advantage-related outcome was coded (sustainable competitive advantage before competitive advantage, firm performance and innovation); new product performance was coded as an innovation outcome. Evidence on P1 was coded only for these four outcomes, and a study whose abstract did not report the direction of an effect was coded as providing insufficient evidence rather than as supportive.

To check the consistency of the coding, all 109 studies were coded a second time, independently of the first coding and on the basis of a written codebook that defines every category and decision rule. The second coding was carried out with the assistance of a large language model that applied the codebook to the abstracts without access to the first coding. Agreement between the two codings was high for tier (κ = 0.90), research design (κ = 0.84) and outcome type (κ = 0.84), substantial for P1 (κ = 0.70) and moderate for P2 (κ = 0.53) and for sensing, seizing and transforming (κ = 0.37 to 0.47); for the quality criteria Q1 to Q6, the two codings agreed in 80 to 99 per cent of the cases. All disagreements were resolved by re-applying the codebook, and the adjudicated codes form the basis of Chapter 4. The lower agreement for the three dynamic capabilities shows that their assignment involves interpretation, which is why the corresponding results are reported as tendencies rather than exact frequencies.

In addition, the full text of every open-access study was checked where it could be retrieved. Of the 52 open-access studies identified first, 29 full texts were accessible; a later search via Unpaywall and OpenAlex yielded eight further full texts, partly as accepted manuscripts or preprints in institutional repositories. The 37 full texts were used to verify the coding of the outcome, of the direct effect and of the sample, and to assess whether limitations are discussed; for the eight later studies, the dynamic capabilities addressed and the quality criteria were also re-checked. For each study, the data extraction sheet records whether the coding is based on the full text or on the abstract only. All codes were recorded together with a short summary of the main findings, the mediators and the moderators reported in each study, so that every statement in Chapter 4 can be traced back to the individual studies on which it is based. For the 72 studies coded from abstracts, details that are only reported in the full texts, such as measurement instruments or robustness tests, could not be considered systematically. This limitation is discussed in Section 3.6.

3.5 Quality Assessment and Evidence Synthesis

Since the included studies differ considerably in design and rigour, their quality is assessed with a checklist of seven criteria, shown in Table 3.4. Criteria Q1 to Q6 are rated as met (1) or not met (0), resulting in a quality score between 0 and 6. Studies

that meet all six criteria are classified as high quality, studies with 4 or 5 as medium quality and studies with 3 or less as low quality. Criterion Q7 is rated as yes, no or not assessable and is reported separately from the score, as explained below. Because the criteria have to be applicable to quantitative, qualitative, modelling and conceptual studies, they refer to general characteristics of good research rather than to design-specific standards. The score is based only on the methodology and evidence of a study; the journal's VHB listing is reported separately and does not enter the score. The quality assessment is not used to exclude further studies; instead, it determines how much weight a study receives in the synthesis, so that conclusions rest primarily on high-quality evidence.

TABLE 3.4: Study Quality Assessment Framework

No.	Assessment question	Rating
Q1	Is the research aim or question clearly stated?	1 = yes / 0 = no
Q2	Is the study explicitly grounded in theory (e.g. RBV or dynamic capabilities)?	1 / 0
Q3	Is the research design appropriate for the aim?	1 / 0
Q4	Are the sample, data or cases described transparently?	1 / 0
Q5	Is the analysis rigorous (e.g. validity and reliability tests, systematic coding, explicit model assumptions)?	1 / 0
Q6	Are the conclusions supported by the data or arguments presented?	1 / 0
	Quality score Q1–Q6 (high: 6; medium: 4–5; low: 0–3)	0–6
Q7	Are the limitations of the study discussed? (reported separately)	yes / no / not assessable
	Recorded separately (not scored): journal listed in the VHB Rating 2024	yes / no

Source: Author's own work

Some criteria require interpretation for particular designs. For conceptual papers, criterion Q4 is interpreted as the transparency of the conceptual approach, for example an explicit description of the theories and logic used to derive propositions, and criterion Q5 as the rigour of the argumentation. For formal models, Q4 refers to the explicit statement of model assumptions. Criterion Q7 requires particular care. Abstracts rarely mention limitations, so rating Q7 as not met whenever an abstract is silent would systematically underrate all studies coded from abstracts and would distort the comparison with studies whose full text was read. Q7 was therefore rated as yes or no only where the full text was checked or the abstract itself refers to limitations, and as not assessable otherwise. Because Q7 cannot be assessed for most studies, it is reported separately and does not enter the quality score, so that the score is based on the same six criteria for every study.

The evidence is synthesised thematically. Thematic synthesis was developed to bring together and integrate the findings of multiple studies within systematic reviews (Thomas & Harden, 2008, p. 1), and thematic analysis is regarded as an accessible and theoretically flexible approach to identifying patterns in qualitative material (Braun & Clarke, 2006, p. 77). In this thesis, the extracted findings are first coded according to the elements of the conceptual framework: AI capabilities, sensing, seizing, transforming, competitive advantage, sustainable competitive advantage and contextual factors. These codes are then grouped into themes that correspond to the research questions, while new themes that emerge from the data are added where necessary. The results are reported in Chapter 4 along these themes.

Finally, the propositions are evaluated according to the rules defined in Section 1.4. A proposition is considered supported if the majority of the relevant studies, and especially the high-quality studies, report consistent findings; contradicted if the evidence predominantly points in the opposite direction; and inconclusive if the findings are mixed or the number of relevant studies is too small. Evidence from core studies in e-commerce and platform contexts is reported separately from the evidence of supporting studies, so that findings from other industries are not mistakenly presented as context-specific evidence.

3.6 Methodological Limitations

Several limitations of the methodology have to be acknowledged. First, the search was conducted in a single discovery service. Although the EBSCO Discovery Service covers many of the relevant databases and publishers, studies that are only indexed elsewhere may have been missed. The inclusion of 16 studies through other methods partly compensates for this, but it also shows that keyword-based searches do not capture all relevant studies, since authors use different terms for related concepts.

Second, the screening and eligibility assessment were based on titles, abstracts, keywords and bibliographic information and were carried out by a single reviewer. Abstract-based decisions may misclassify some studies, and a single reviewer cannot provide the inter-rater reliability that a team of reviewers would offer. To reduce this risk, explicit criteria and reason codes were defined in advance and all decisions were documented in the screening log, so that they can be reviewed and replicated. The data extraction was checked through an independent, AI-assisted second coding and a full-text check of the accessible open-access studies (Section 3.4), but 72 of the 109 studies are still coded from their abstracts only.

Third, the restriction to English-language, peer-reviewed journal articles excludes conference papers, books and studies in other languages, which may introduce a language and publication bias. Fourth, because journal rankings were not used as an inclusion criterion, the sample contains studies of very different quality, including studies from journals with less rigorous review processes. This increases the coverage of the evidence, but it requires that the quality assessment is applied consistently, and conclusions that rest mainly on low-quality studies must be interpreted with caution. Finally, the search covers publications up to October 2026, and a considerable share of the records are very recent. Of the 58 studies dated 2026, the publication records show that 32 appeared in a 2026 journal issue, 18 have been assigned to a volume without a print date yet, three were published online ahead of print, and for five the publication details could only be verified through the library record because no DOI record was available. The year of publication was therefore checked for every study, but the bibliographic details of some recent articles may still change. Research on AI is developing rapidly, so the evidence base will continue to change after the completion of this review.

4 Results and Discussion

This chapter presents the results of the systematic literature review. Section 4.1 describes the characteristics of the 109 included studies. Sections 4.2 to 4.4 synthesise the evidence along the research questions: the link between AI capabilities and competitive advantage, the role of sensing, seizing and transforming, and the mediating role of dynamic capabilities. Section 4.5 evaluates the three research propositions, and Section 4.6 discusses the findings in light of the theoretical framework developed in Chapter 2. Unless stated otherwise, the findings refer to the final, adjudicated coding described in Section 3.4, which is based on the abstracts and, for the 29 studies whose full text could be accessed, on the full text.

4.1 Literature Search and Study Characteristics

The review includes 109 studies, which are listed individually in Appendix B. Table 4.1 summarises their main characteristics. The most striking feature of the evidence base is its recency. Only 15 studies were published between 2020 and 2023, whereas 13 appeared in 2024, 23 in 2025 and 58 in 2026, including articles published online ahead of print. Almost three quarters of the evidence is therefore less than two years old. This confirms the observation in Chapter 2 that research on AI capabilities is a young and rapidly growing field, but it also means that many findings have not yet been replicated or debated in subsequent work.

Of the included studies, 18 are core studies set in e-commerce, online retail or platform-based businesses, and 91 are supporting studies from other industries. The supporting studies cover a wide range of contexts, most frequently manufacturing, small and medium-sized enterprises, banking and financial services, hospitality, supply chains and start-ups. The predominance of supporting studies underlines that the e-commerce-specific evidence on AI capabilities and competitive advantage is still limited, which is consistent with research gap G1 (Table 2.3).

TABLE 4.1: Publication Characteristics of Included Studies

Characteristic	Category	n
Publication year	2020–2023 / 2024 / 2025 / 2026	15 / 13 / 23 / 58
Tier	Core (e-commerce, online retail, platforms) / supporting	18 / 91
Research design	Survey (mostly PLS-SEM)	56
	Conceptual	16
	Mixed methods	15
	Qualitative and case studies	13
	Panel or archival data	7
	Formal (game-theoretic) models	2
Region (where reported; multiple possible)	China	17
	South and South-East Asia (e.g. India, Pakistan, Vietnam)	14
	Middle East and North Africa (e.g. Saudi Arabia, Jordan, Egypt)	13
	Europe	8
	North America	5
	Other (e.g. Taiwan, Hong Kong, Australia, multi-country)	7
	Not reported in the abstract	47
Theoretical lens	Dynamic capabilities / RBV / both / none named	56 / 27 / 14 / 30
Journal	Listed in the VHB Rating 2024	56
Quality (Table 3.4, Q1–Q6)	High (6) / medium (4–5) / low (0–3)	53 / 45 / 11
Limitations discussed (Q7)	Yes / no / not assessable	34 / 8 / 67
Evidence source	Full text verified / abstract only	37 / 72

Source: Author's own work based on the data extraction

Methodologically, the evidence is dominated by cross-sectional surveys of managers, most of them analysed with partial least squares structural equation modelling. Conceptual studies form the second largest group, followed by qualitative case studies. Panel studies based on archival data are rare and are concentrated in China (Liu & Xin, 2026, p. 1; Huang & Yang, 2026, p. 1; Dai & Zhang, 2026, p. 1). Only a handful of studies use designs that observe firms over time, such as a three-wave survey of Vietnamese start-ups (Le et al., 2026, p. 1) or a time-lagged survey in Saudi retail and e-commerce (Alnahdi, 2026, p. 1). Geographically, the evidence comes mainly from China, the Middle East and South Asia. Only eight studies refer to European firms, and

none of the studies that report their context is set in Germany. For a thesis written in Germany, this is an important limitation of the transferability of the findings, which is taken up again in Section 4.6.

Dynamic capabilities theory is the most frequently used theoretical lens: 56 studies refer to it explicitly, 27 to the resource-based view and 14 combine both perspectives. At the same time, 30 studies do not name a theoretical foundation in their abstract. According to the quality assessment on criteria Q1 to Q6, 53 studies were rated as high quality, 45 as medium quality and 11 as low quality, with a mean score of 5.1 out of 6. Whether limitations are discussed (Q7) could be assessed for only 42 studies: 34 of them discuss their limitations and eight do not, while for the remaining 67 studies this criterion is not assessable from the abstract and is therefore reported separately rather than scored as zero. Quality is not related to the journal listing: among the 56 studies in VHB-listed journals, 26 were rated as high quality, compared with 27 of the 53 studies in non-listed journals. The studies are spread across a large number of journals, and no single outlet dominates. The most frequent outlets are *Strategic Business Research* (seven studies), *Industrial Marketing Management* and the *International Journal of Information Management* (five each), the *Journal of Business Research* (four) and the *Academy of Management Review* (three). This dispersion reflects the interdisciplinary nature of the topic, which connects strategic management, information systems, marketing and operations research.

Figure 4.1 shows how the studies are distributed across outcome types. Firm performance is the most frequently studied outcome (32 studies), followed by competitive advantage (23 studies). Firm performance was coded separately and was not treated as competitive advantage, and competitive advantage was only coded as sustainable when the authors explicitly framed it as sustained or long-term. On this basis, only 11 studies examine sustainable competitive advantage or a comparable long-term outcome, and none of these is a core study from e-commerce or platform contexts. Further studies address innovation outcomes, including new product performance (11), resilience (8) and business model innovation (6), while 18 studies examine other outcomes such as strategic renewal, user value or implementation success.

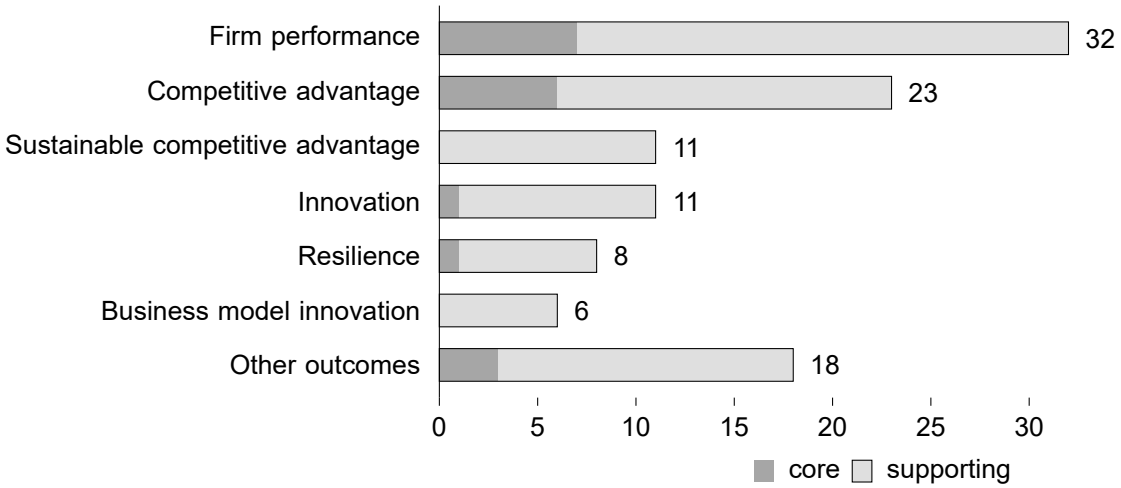


FIGURE 4.1: Distribution of Studies by Research Theme

Source: Author's own work based on the data extraction (n = 109)

4.2 AI Capabilities and Competitive Advantage

The first research question asks how AI capabilities are associated with competitive advantage. Of the 109 studies, 79 provide evidence on this link. In 63 of them, AI capabilities are positively associated with competitive advantage, firm performance or innovation outcomes, nine studies describe the relationship as conditional, five find no significant direct effect and two highlight constraints that prevent firms from turning AI into an advantage. For the remaining 30 studies the evidence is insufficient, either because they examine other outcomes or, in one case, because the abstract does not report the direction of the effect (Chatterjee et al., 2021, p. 205). Table 4.2 summarises this evidence by outcome type.

TABLE 4.2: Evidence Linking AI Capabilities to Competitive Advantage

Outcome type	n	Direction of evidence	Representative studies
Firm performance	32	29 positive, 3 n.s. direct effect	Core: (Chen et al., 2022, p. 1; Alnahdi, 2026, p. 1; Sivathanu & Pillai, 2026, p. 93); supporting: (Mikalef & Gupta, 2021, p. 1; Fosso Wamba, 2022, p. 1; Ayoub & Sopuru, 2026, p. 1; Kwiatek, 2025, p. 1)
Competitive advantage	23	16 positive, 5 conditional, 1 constraint, 1 insufficient	Core: (Bunea & Corboș, 2026, p. 1; Awamleh & Bustami, 2022, p. 1; Guo et al., 2025, p. 1250; Chen et al., 2026, p. 187); supporting: (Khattab et al., 2026, p. 1; Sukati et al., 2026, p. 1; Li et al., 2025, p. 10; Krakowski et al., 2023, p. 1425)
Sustainable competitive advantage	11	6 positive, 2 n.s., 2 conditional, 1 constraint	Supporting only: (Liang & Sun, 2026, p. 1; Hossain et al., 2022, p. 240; Alioune, 2026, p. 1; Wu et al., 2026, p. 1; McIlveene et al., 2026, p. 1)
Innovation (incl. new product performance)	11	10 positive, 1 conditional	(Kyriakopoulos et al., 2025, p. 1; Dinh, 2026, p. 84; Tao et al., 2026, p. 44; Liu & Xin, 2026, p. 1)
Resilience, business models	14	Link to advantage rarely tested	(Dai & Zhang, 2026, p. 1; AlQershi, 2026, p. 1; Sjödin et al., 2021, p. 574; Zhong, 2026, p. 1)

Source: Author's own work based on the data extraction; n.s. = not significant

The core studies from e-commerce consistently report positive associations, but mostly with performance rather than with competitive advantage in the strict sense. In a survey of 394 e-commerce firms, AI capability affects firm performance only indirectly, through creativity, AI management and AI-driven decision making (Chen et al., 2022, p. 1). In Saudi retail and e-commerce, AI capability contributes to organisational performance by facilitating service innovation and improved consumer outcomes, with digital infrastructure acting as a supporting factor (Alnahdi, 2026, p. 1). A survey of e-commerce managers examines how the use of generative AI builds technological capabilities that are linked to financial performance in terms of sales, revenue and profitability (Sivathanu & Pillai, 2026, p. 93), and in Saudi SMEs the adoption of AI in e-commerce enhances business performance (Aljarboa, 2024, p. 1). Operational benefits are reported as well: AI usage improves the logistics efficiency of Indian e-commerce businesses, partly through supply chain consistency and last-mile delivery performance (Malhotra & Kharub, 2025, p. 290), and qualitative evidence from Philippine clothing retailers shows that even free AI tools improve management, targeted

advertising and customer communication (Bernardo, 2026, p. 76).

Further core studies extend this picture to marketing and to cross-functional effects. In Egyptian e-retail stores, AI applications are strongly correlated with marketing innovation, with organisational culture acting as a mediator (Awad, 2024, p. 170). A survey of 750 business managers, financial executives and e-commerce practitioners conceptualises AI adoption as a cross-functional capability and finds consistent positive effects on business management, e-commerce and finance (Haqqani & Aleem, 2026b, p. 1). A literature-based conceptual study describes personalisation, predictive analytics and automation as the core AI capabilities that transform marketing in e-business environments, with customer engagement and satisfaction linking these capabilities to marketing outcomes (Mittal, 2025, p. 1). These studies confirm that, in e-commerce, the benefits of AI are closely tied to customer-facing processes, but they also rely on correlational or conceptual evidence.

Only a few core studies measure competitive advantage directly. In electronic retailing, AI implementation is associated with qualitative and quantitative benefits that are linked, through customer experience, to differentiation- and cost-based competitive advantage (Bunea & Corboș, 2026, p. 1). Among Jordanian e-commerce firms, IT capabilities mediate the relationship between AI and competitive advantage (Awamleh & Bustami, 2022, p. 1), and for Chinese SMEs on cross-border platforms, the coupling of AI adoption and strategic agility improves their position in global value chains through international marketing capabilities (Guo et al., 2025, p. 1250). Platform-specific theory points in the same direction but emphasises dynamics: the AI capability of a platform increases the value users perceive through data network effects (Gregory et al., 2021, p. 534), and an entrant with a superior algorithm can overcome an incumbent's user base (Chen et al., 2026, p. 187). A game-theoretic study of live streaming shows, however, that the competitive advantage of an online retailer depends on conditions such as online acceptance and the relative advantage of human-hosted or AI-supported streaming (Zhou et al., 2025, p. 1348).

Two further core studies and one related supporting study conceptualise how AI creates advantage at the level of platforms and ecosystems rather than individual firms. One framework explains how AI-based capabilities are translated into dynamic competitive positioning through strategic intelligence, the orchestration of platform ecosystems and network effects (Ratna, 2026, p. 649). In solution-based business models of manufacturers, AI platforms are argued to foster collaborative value systems between firms and other value-chain actors across problem identification, solution development and implementation, and thereby to support the development of situating capabilities (Vaillant

et al., 2025, p. 1). At the level of multinational enterprises, orchestration theory proposes that firms create value by orchestrating globally distributed resources and partners within digital platform and AI ecosystems (Teece & Pitelis, 2026, p. 689). These contributions shift the unit of analysis from the firm to the ecosystem: in platform-based business, AI capabilities may create advantage less through internal efficiency than through the ability to coordinate external actors and to learn from the data they generate. Since these studies are conceptual or do not test competitive advantage directly, however, these arguments still await empirical testing.

The supporting studies confirm the positive association across many industries. AI capabilities increase competitive advantage in Jordanian manufacturing firms, partly through organisational agility (Khattab et al., 2026, p. 1), in micro, small and medium-sized enterprises through supply chain agility and resilience (Sukati et al., 2026, p. 1), and in B2B firms through service innovation (Li et al., 2025, p. 10). Comparable results are reported for hotels (Elshaer et al., 2025, p. 1), Vietnamese SMEs (Giang et al., 2026, p. 1), multinational corporations (Zhang, 2026, p. 1) and Tunisian family firms (Gafsi, 2026, p. 1). In customer-facing functions, one study examines how AI-based CRM relates to performance and competitive advantage depending on firm size, age and industry, although its abstract does not report the direction of the effects (Chatterjee et al., 2021, p. 205), and an AI-powered CRM capability improves profitability and competitive advantage through marketing ambidexterity (Alnofeli et al., 2026, p. 1).

Across industries, the supporting studies reveal recurring patterns that are relevant for e-commerce. In manufacturing, AI capabilities are mainly linked to innovation and to new service-based business models: they foster servitization, especially when absorptive capacity is high (Abou-Foul et al., 2023, p. 1), improve the novelty and efficiency of business model design (Zhong, 2026, p. 1) and support product and new product performance (Gao et al., 2025b, p. 1; Tao et al., 2026, p. 44). In hospitality, AI capabilities create advantage through supply chain agility and resilience (Elshaer et al., 2025, p. 1) and through co-innovation (Naz et al., 2024, p. 2055). In banking and financial services, AI-enabled risk analytics (Salem et al., 2026, p. 1) and AI-powered CRM (Alnofeli et al., 2026, p. 1) are linked to financial performance and competitive advantage. Studies of SMEs and start-ups emphasise agility and business model change, but also resource constraints (Giang et al., 2026, p. 1; Wishah et al., 2026, p. 1; Le et al., 2026, p. 1). Several of these mechanisms, in particular customer relationship management, service innovation, supply chain agility and business model change, are central to e-commerce as well. This supports the transferability of the supporting evidence, while the specific role of data network effects and platform competition can only be assessed from the core studies.

The 11 studies that address sustainable competitive advantage deserve particular attention, because this outcome is at the centre of the thesis. Six of them are conceptual. One proposes that AI-driven strategic capabilities, such as AI-enabled market intelligence and AI governance, lead to sustainable competitive advantage through an organisational sustainability capability that enables firms to sense, seize and realign their resources (Htet et al., 2026, p. 1). Another links AI-driven capabilities, embedded sustainability and adaptive organisational cultures to durable advantage (Wei, 2026, p. 1), and a third develops a framework for AI-driven branding aimed at sustained brand advantage (Deryl et al., 2025, p. 1). Empirically, a marketing analytics capability helps exporters sense, seize and reconfigure the market, leading to a sustained competitive advantage, and AI adoption strengthens these capabilities (Hossain et al., 2022, p. 240). Panel data from Chinese listed firms show that AI helps firms balance short-term returns and long-term growth through organisational resilience and strategic change (Liang & Sun, 2026, p. 1), and AI capability maturity is linked to competitive performance framed as sustained advantage through human-AI integration and marketing agility (Sharma, 2026, p. 1). In contrast, the direct effect of AI capabilities on sustainable competitive advantage is not significant in Spanish high-growth industries (Alioune, 2026, p. 1), the full text of a study of Chinese high-tech firms shows that AI capability neither has a significant main effect on sustainable competitive performance nor moderates the effect of disruptive innovation (Wang et al., 2026, p. 1), democratised generative AI is argued to yield a sustained advantage only for firms that build proprietary perceptual systems (Cui et al., 2024, p. 583), small businesses often fail to achieve a lasting advantage from AI (McIlveene et al., 2026, p. 1), and whether AI produces fleeting efficiency gains or sustained advantages depends on how managers frame it (Wu et al., 2026, p. 1). The evidence on sustainability is therefore thin, comes exclusively from supporting studies and relies mostly on conceptual reasoning or on managers' perceptions rather than on observed persistence of advantage.

A closer look at the remaining studies reveals that a direct effect of AI capabilities on competitive outcomes is not the rule. In five studies, the direct link is not significant. In Spanish high-growth industries, digital transformation leadership affects sustainable competitive advantage, whereas the direct effect of AI capabilities is not significant (Alioune, 2026, p. 1), and in Chinese high-tech firms AI capability has no significant main effect on sustainable competitive performance (Wang et al., 2026, p. 1). A reanalysis of SME data finds no significant direct link between AI capability and performance, but a positive indirect effect through AI adoption and dynamic capabilities (Akaelu et al., 2026, p. 1). In early-stage ventures, strategic resilience fully accounts for the relationship between AI capability and entrepreneurial success (Wang & Zhang, 2026, p. 1), and in B2B loyalty programmes, the direct effect of AI capability on performance is weak

and not significant, while three quarters of the total effect run through customer-relating capability and programme adaptation (Kwiatek, 2025, p. 1). Several further studies report positive effects only as indirect or serial mediation effects (Haqqani & Aleem, 2026a, p. 1; Sharma, 2026, p. 1).

Finally, a smaller group of studies questions whether AI can be a source of advantage at all. In a controlled setting, AI adoption makes traditional human capabilities obsolete while creating new human-machine capabilities, thereby shifting the sources of competitive advantage (Krakowski et al., 2023, p. 1425). Conceptual work argues that AI only becomes a source of advantage when it is "situated" in firm-specific activities of grounding, bounding and recasting (Kemp, 2024, p. 618), and that democratised generative AI provides generic insights unless firms build proprietary perceptual systems (Cui et al., 2024, p. 583). Small businesses often fail to turn AI into a lasting advantage because of gaps in financial, human, technological and organisational resources (McIlveene et al., 2026, p. 1), and some firms even exaggerate their AI capabilities to gain an apparent advantage, which puts trust and reputation at risk (Elhajjar & Itani, 2025, p. 519). Evidence from strategic decision making suggests that large language models can already generate and evaluate strategies at a level comparable to entrepreneurs and investors, but that the effect on firm performance will depend on competitive dynamics as these capabilities diffuse (Csaszar et al., 2024, p. 322).

4.3 AI Capabilities and Dynamic Capabilities

The second research question concerns the relationship between AI capabilities and dynamic capabilities. This relationship is addressed by 79 studies, and 74 of them report that AI capabilities strengthen dynamic capabilities or that dynamic capabilities explain how AI capabilities create value. Two studies find that the tested capability does not mediate the effect of AI on performance (Kokash, 2026, p. 1; Akter et al., 2026, p. 1), and three describe the role of dynamic capabilities as conditional, for example when AI capabilities only moderate another capability path (Wang et al., 2026, p. 1; Hossain et al., 2022, p. 240; Kemp, 2024, p. 618). Table 4.3 summarises the evidence for each of the three dynamic capabilities.

TABLE 4.3: Evidence on Sensing, Seizing and Transforming

Capability	n	AI-enabled mechanisms identified	Representative studies
Sensing	42	Competitive and risk intelligence, market exploration, pattern recognition, information management, strategy search	(Maune, 2026, p. 6; AlQershi, 2026, p. 1; Kumar et al., 2025b, p. 1; Csaszar et al., 2024, p. 322)
Seizing	73	Organisational and strategic agility, decision-making agility, service and marketing innovation, business model design	(Kyriakopoulos et al., 2025, p. 1; Khattab et al., 2026, p. 1; Zhong, 2026, p. 1; Li et al., 2025, p. 10)
Transforming	46	Business model innovation and transformation, resource orchestration, strategic renewal, reconfiguration	(Sjödin et al., 2021, p. 574; Le et al., 2026, p. 1; Liu & Xin, 2026, p. 1; Huang & Yang, 2026, p. 1)
All three	24	AI capabilities mapped explicitly on sensing, seizing and transforming	(Kao, 2026, p. 109; Chu & Chan, 2026, p. 1; Mikalef et al., 2021, p. 80; Hossain et al., 2022, p. 240)

Source: Author's own work based on the data extraction; studies can address several capabilities

Sensing. Forty-two studies describe how AI improves the ability to identify opportunities and threats. AI capability strongly predicts the effectiveness of competitive intelligence, which in turn drives corporate growth (Maune, 2026, p. 6). Multinational enterprises with advanced AI-driven risk intelligence are more resilient to geopolitical shocks, adapting their supply chains more rapidly (AlQershi, 2026, p. 1), and AI-driven risk management is associated with supply chain resilience in the energy sector (Alhamaideh et al., 2026, p. 1). In international marketing, AI capabilities enable firms to explore new export markets (Kumar et al., 2025b, p. 1), and in financial institutions they improve information management and planning (Bhatti & Pardaev, 2026, p. 1). At the level of strategy itself, large language models can support search and the evaluation of strategic alternatives (Csaszar et al., 2024, p. 322).

Although sensing is the dynamic capability that AI appears most naturally suited to support, it is studied far less often than seizing. Moreover, the studies indicate that better information does not automatically lead to better action. Multinational enterprises show substantial heterogeneity in the value they derive from AI because the translation from insight to action is hindered by governance challenges, such as the contested

legitimacy of algorithmic recommendations and unclear accountability (Chu & Chan, 2026, p. 1). Similarly, the effect of AI depends on whether managers interpret it as a tool, an actor or a strategic resource (Wu et al., 2026, p. 1). AI-enabled sensing therefore only contributes to competitive advantage if it is connected to the seizing and transforming activities through which firms act on new information.

Seizing. Seizing is by far the most frequently addressed capability (73 studies). The most common mechanism is agility: AI capabilities increase new product development decision-making agility (Kyriakopoulos et al., 2025, p. 1), organisational agility (Khattab et al., 2026, p. 1; Giang et al., 2026, p. 1), strategic agility (Wang et al., 2025, p. 3003) and supply chain agility (Sukati et al., 2026, p. 1; Elshaer et al., 2025, p. 1), and AI assimilation improves performance through organisational and customer agility (Fosso Wamba, 2022, p. 1). A second mechanism is the design of new offerings and business models: AI capability improves both the novelty and the efficiency of business model design (Zhong, 2026, p. 1), generative AI capabilities foster service innovation (Li et al., 2025, p. 10), and AI capabilities strengthen marketing capabilities (Kumar et al., 2025a, p. 1).

Transforming. Forty-three studies address the transformation or reconfiguration of the resource base. Scaling AI requires manufacturers to innovate their business models through agile customer co-creation, data-driven delivery operations and ecosystem integration (Sjödin et al., 2021, p. 574), and AI adoption and capabilities enhance business model transformation in start-ups (Le et al., 2026, p. 1). Dynamic capability amplifies the innovation returns to AI, indicating larger gains when firms can orchestrate and reconfigure resources (Liu & Xin, 2026, p. 1). The panel study of Chinese listed firms is particularly instructive: sensing and seizing primarily strengthen efficiency-oriented value propositions, whereas reconfiguring capability is required for product-oriented value propositions (Huang & Yang, 2026, p. 1). At the same time, transformation is demanding: relatively few established organisations have fundamentally transformed their business models through AI (Black et al., 2024, p. 1), and internal organisational dynamics rather than AI-induced market pressure drive strategic renewal (Bughin, 2026, p. 1242).

A further observation concerns the way in which AI capabilities are conceptualised. Some studies treat AI capability as a higher-order construct composed of AI-specific resources (Mikalef & Gupta, 2021, p. 1), of basic resources, proclivity and skills (Chen et al., 2022, p. 1), of four sub-dimensions (Abou-Foul et al., 2023, p. 1) or of automation, insights and personalisation (Kwiatek, 2025, p. 1). Others focus on specific functional applications, such as AI-based CRM (Chatterjee et al., 2021, p. 205; Alnofeli et al., 2026, p. 1), AI-enabled risk analytics (Salem et al., 2026, p. 1; Alhamaideh et al., 2026,

p. 1) or AI-driven supply chain capabilities (Sukati et al., 2026, p. 1; Elshaer et al., 2025, p. 1). A growing group examines generative AI capabilities (Li et al., 2025, p. 10; Qiao et al., 2026, p. 7274; Wishah et al., 2026, p. 1; Ruangkanjanases et al., 2025, p. 1; Sivathanu & Pillai, 2026, p. 93), and the most recent work already discusses agentic AI (Progga & Ali, 2026, p. 1). This heterogeneity mirrors the underspecification of the dynamic capabilities construct noted by Schilke et al. (2018, p. 390) and makes it difficult to compare effect sizes across studies. It is also one reason why the synthesis in this chapter focuses on the direction and nature of relationships rather than on their magnitude.

The relationship between AI and dynamic capabilities is not only one-directional. Several studies conceptualise AI-enabled capabilities as dynamic capabilities in their own right, for example AI-enabled dynamic capability (Ayoub & Sopuru, 2026, p. 1), AI-enabled knowledge capabilities (Dinh, 2026, p. 84) or AI orchestration capability as a “synthetic” dynamic capability that coordinates data, models and decision authority (Chu & Chan, 2026, p. 1). Case evidence identifies AI-specific microfoundations of dynamic capabilities (Mikalef et al., 2021, p. 80) and six AI capabilities linked to sensing, seizing and reconfiguring in the pursuit of business sustainability (Kao, 2026, p. 109). In addition, the development of AI-driven dynamic capabilities depends on entrepreneurial orientation (Schneider et al., 2026, p. 1) and on the dynamic managerial capabilities that translate into AI integration capability (Li et al., 2026, p. 1).

4.4 The Mediating Role of Dynamic Capabilities

The results so far suggest that dynamic capabilities, and capabilities closely related to them, act as the main mechanism through which AI capabilities affect competitive outcomes. Table 4.4 groups the mediators reported in the included studies into seven families. Innovation-related capabilities, including business model innovation and ambidexterity, are the most frequent mediators (16 studies), followed by IT, data and technological capabilities (11), agility (10) and marketing or customer-related capabilities (10).

TABLE 4.4: Summary of Mediation Evidence

Mediator family	n	Representative studies
Innovation, business model innovation, ambidexterity	16	(Li et al., 2025, p. 10; Wishah et al., 2026, p. 1; Kumar et al., 2025b, p. 1; Alnofeli et al., 2026, p. 1)
IT, data and technological capabilities	11	(Sivathanu & Pillai, 2026, p. 93; Awamleh & Bustami, 2022, p. 1; Ayoub & Sopuru, 2026, p. 1; Gao et al., 2025b, p. 1)
Agility (organisational, strategic, supply chain)	10	(Khattab et al., 2026, p. 1; Giang et al., 2026, p. 1; Elshaer et al., 2025, p. 1; Fosso Wamba, 2022, p. 1)
Marketing and customer-related capabilities	10	(Kwiatek, 2025, p. 1; Kumar et al., 2025a, p. 1; Guo et al., 2025, p. 1250; Bunea & Corboș, 2026, p. 1)
Decision making and cognition	8	(Haqqani & Aleem, 2026a, p. 1; Maune, 2026, p. 6; Chen et al., 2022, p. 1)
Resilience (strategic, organisational, supply chain)	7	(Wang & Zhang, 2026, p. 1; Liang & Sun, 2026, p. 1; Sukati et al., 2026, p. 1)
Dynamic capabilities in general	7	(Akaelu et al., 2026, p. 1; Qiao et al., 2026, p. 7274; Liu & Xin, 2026, p. 1)
Full or predominant mediation: (Wang & Zhang, 2026, p. 1; Kwiatek, 2025, p. 1; Akaelu et al., 2026, p. 1; Chen et al., 2022, p. 1); *partial mediation:* (Fosso Wamba, 2022, p. 1; Bunea & Corboș, 2026, p. 1; Dinh, 2026, p. 84; Ayoub & Sopuru, 2026, p. 1); *no mediation:* (Kokash, 2026, p. 1; Akter et al., 2026, p. 1)		

Source: Author's own work based on the data extraction; studies can report several mediators

The strength of mediation varies. In four studies, the effect of AI capabilities on outcomes runs entirely or predominantly through mediating capabilities: strategic resilience (Wang & Zhang, 2026, p. 1), customer-relating capability and programme adaptation (Kwiatek, 2025, p. 1), AI adoption and dynamic capabilities (Akaelu et al., 2026, p. 1), and creativity, AI management and AI-driven decision making (Chen et al., 2022, p. 1). In other studies, the mediation is partial: AI assimilation improves performance directly and through organisational and customer agility, which act as complementary partial mediators (Fosso Wamba, 2022, p. 1); customer experience partially mediates the effects of AI-enabled benefits on competitive advantage in electronic retailing (Bunea & Corboș, 2026, p. 1); and AI-enabled knowledge capabilities affect new product performance both directly and through innovation radicalness (Dinh, 2026, p. 84). Two counter-examples stand out: in tourism, AI capabilities directly improve firm performance and dynamic capabilities do not mediate this relationship (Kokash, 2026, p. 1), and in an entrepreneurial setting innovation capability does not mediate the effect of AI usage on performance (Akter et al., 2026, p. 1).

Several studies identify sequential chains of capabilities. AI capability maturity is associated with human-AI integration, which is linked to marketing agility and, in turn, to competitive performance (Sharma, 2026, p. 1). In Chinese listed firms, organisational resilience and strategic change act as parallel and chain mediators between AI and the balance of short-term returns and long-term growth (Liang & Sun, 2026, p. 1). Such chains are consistent with the hierarchical view of capabilities, in which AI capabilities form a lower-order capability that has to be transformed into higher-order dynamic and operational capabilities before it creates value.

The mediating mechanisms are themselves contingent, as several moderated mediation studies show. In Jordanian SMEs, market sensing agility strengthens the path from generative AI capability through business model innovation to performance (Wishah et al., 2026, p. 1). Conversely, environmental dynamism weakens the mediating role of service concept and customer experience innovation between generative AI capabilities and competitive advantage (Li et al., 2025, p. 10). Socio-cognitive sensemaking strengthens the mediation through digital resilience in the case of digital product innovation (LI et al., 2025, p. 1), competitive pressure strengthens the effects of AI-driven supply chain agility and resilience (Sukati et al., 2026, p. 1; Elshaer et al., 2025, p. 1), and knowledge governance mechanisms moderate the effect of AI capability on business intelligence transforming capability (Gao et al., 2025b, p. 1). These findings link the second and third research questions: the dynamic capabilities mechanism is not a fixed pathway but depends on managerial cognition, governance and the competitive environment.

Resilience deserves particular attention among the mediators, because it is the outcome most closely related to the persistence of advantage. AI innovation capability creates entrepreneurial success only through strategic resilience (Wang & Zhang, 2026, p. 1), organisational resilience and strategic change link AI to the balance of short-term returns and long-term growth (Liang & Sun, 2026, p. 1), and AI adoption improves supply chain resilience through transparency, efficiency and optimised supplier concentration (Dai & Zhang, 2026, p. 1). Firms with AI-driven risk intelligence are also more resilient to geopolitical shocks (AlQershi, 2026, p. 1). Since resilience describes the capacity to sustain functioning under disruption, it may represent a missing link between AI-enabled dynamic capabilities and sustainable competitive advantage, a link that has not yet been tested directly in e-commerce.

From the perspective of this thesis, it is important that these mediators are not limited to dynamic capabilities in the narrow sense. Agility and resilience are closely related to sensing and seizing, while business model innovation and reconfiguration correspond to transforming. Other mediators, such as marketing capabilities or IT capabilities, are

operational capabilities that dynamic capabilities help to create and renew. The evidence therefore supports a broad interpretation of the dynamic capabilities mechanism: AI capabilities create value when they are converted into the ability to adapt, innovate and reconfigure, rather than through their mere presence.

4.5 Evaluation of Research Propositions

The three propositions formulated in Section 1.4 are evaluated according to the rules defined in Section 3.5. Table 4.5 summarises the assessment.

TABLE 4.5: Research Proposition Evaluation

Proposition	Evidence	Assessment
P1: AI capabilities are associated with competitive advantage	63 of 79 relevant studies positive (29 of 38 high-quality studies); 9 conditional, 5 n.s. direct effect, 2 constraints; 30 insufficient evidence; core studies: 12 of 14 positive, 2 conditional, mostly on performance	Supported, with qualifications (mostly indirect; little evidence on sustainability)
P2: Dynamic capabilities explain how AI capabilities are converted into competitive advantage	74 of 79 relevant studies supportive (39 of 41 high-quality studies); 3 conditional, 2 no mediation; core studies: 7 of 7; seizing dominant (73), transforming (46), sensing (42)	Supported
P3: Sustainability depends on organisational and contextual factors	Contingencies reported in 104 studies: integration and complementary assets (48), environmental dynamism (38), data and infrastructure (29), managerial cognition (22), trust, ethics and regulation (20), firm resources (19), platform conditions (8), imitability (6)	Supported for contingency; inconclusive for sustainability over time

Source: Author's own work based on the data extraction

P1 is supported, but with important qualifications. The large majority of studies, and of the high-quality studies, report a positive association between AI capabilities and competitive advantage or firm performance. The positive association also holds for the core studies from e-commerce and platform contexts. However, the association is mostly indirect, and where mediators are included the direct effect often becomes weak or insignificant (Alioune, 2026, p. 1; Akaelu et al., 2026, p. 1; Wang & Zhang, 2026, p. 1;

Kwiatek, 2025, p. 1; Wang et al., 2026, p. 1). Moreover, most studies measure firm performance or competitive advantage at a single point in time, and only 11 studies, none of them from e-commerce, examine sustainable competitive advantage. The evidence therefore supports the association between AI capabilities and advantage, but not yet the claim that this advantage is sustainable.

P2 is clearly supported. Almost all studies that address dynamic capabilities find that AI capabilities strengthen sensing, seizing or transforming, or that these capabilities explain the effect of AI on outcomes. This holds for all core studies that address the link and for nearly all high-quality studies. The evidence is, however, unevenly distributed across the three capabilities. Seizing is addressed far more often (73 studies) than sensing (42) or transforming (46), and only 24 studies examine all three capabilities together. This imbalance is substantively important, because evidence from panel data suggests that transformation is needed to move beyond efficiency-oriented value creation (Huang & Yang, 2026, p. 1).

P3 is supported with respect to the existence of contingencies. Most studies report factors that strengthen or weaken the effects of AI capabilities. The most frequent are the integration of AI into processes and complementary assets, for example absorptive capacity (Abou-Foul et al., 2023, p. 1), formalisation (Kyriakopoulos et al., 2025, p. 1) or supply chain integration (Zhong, 2026, p. 1; Xu et al., 2026, p. 1); environmental dynamism and turbulence, which can both strengthen (Wang et al., 2025, p. 3003) and weaken (Li et al., 2025, p. 10; Le et al., 2026, p. 1) the effects; data quality and digital infrastructure (Kokash, 2026, p. 1; Alnahdi, 2026, p. 1); and firm resources and characteristics (McIlveene et al., 2026, p. 1; Liang & Sun, 2026, p. 1). Since only a few studies follow firms over time, however, the evidence is inconclusive as to which of these factors determine whether an AI-based advantage persists. P3 is therefore assessed as supported regarding contingency, but inconclusive regarding sustainability over time.

The robustness of these assessments differs between the two tiers of evidence. The core studies support P1 and P2 at least as consistently as the supporting studies: 12 of the 14 core studies that address the link report a positive association and two describe it as conditional, and all seven that address dynamic capabilities find them relevant. However, the core evidence base is small and of mixed strength. Eight of the 18 core studies reach a high quality score (Chen et al., 2022, p. 1; Guo et al., 2025, p. 1250; Modgil et al., 2022, p. 1246; Chen et al., 2026, p. 187), eight are published in journals listed in the VHB Rating 2024, and much of the core evidence stems from conceptual papers, formal models and surveys in a small number of countries. The conclusions for e-commerce and platform businesses therefore rest partly on the supporting studies,

whose transferability to these contexts has to be argued rather than assumed. This is done in the following discussion.

To test the robustness of the assessments, the evidence was also examined for three subsets of studies. When only the 93 empirical studies are considered, 57 of the 67 studies that address P1 report a positive association, and 67 of the 71 studies that address P2 are supportive. When the 11 low-quality studies are excluded, the shares remain virtually unchanged (58 of 73 for P1 and 69 of 74 for P2), and the same holds for the 53 high-quality studies (29 of 38 and 39 of 41). The most interesting pattern emerges for the 56 studies in VHB-listed journals: here, 25 of the 35 studies that address P1 report a positive association, while seven describe the relationship as conditional, two emphasise constraints and one finds no significant direct effect. Studies in established, highly ranked journals thus draw a more nuanced picture of the link between AI capabilities and competitive advantage than the evidence base as a whole, whereas their support for P2 is almost unanimous (38 of 40). This strengthens the conclusion that dynamic capabilities are the key mechanism, and it suggests that the direct link between AI and advantage should be treated with caution.

4.6 Discussion and Implications of the Findings

The findings allow a first answer to the central question of this thesis. AI capabilities are associated with competitive advantage, but this association is largely indirect and depends on the conversion of AI capabilities into dynamic and operational capabilities. This is consistent with the theoretical framework developed in Chapter 2 and refines it in four respects, which are summarised in the evidence-informed model in Figure 4.2.

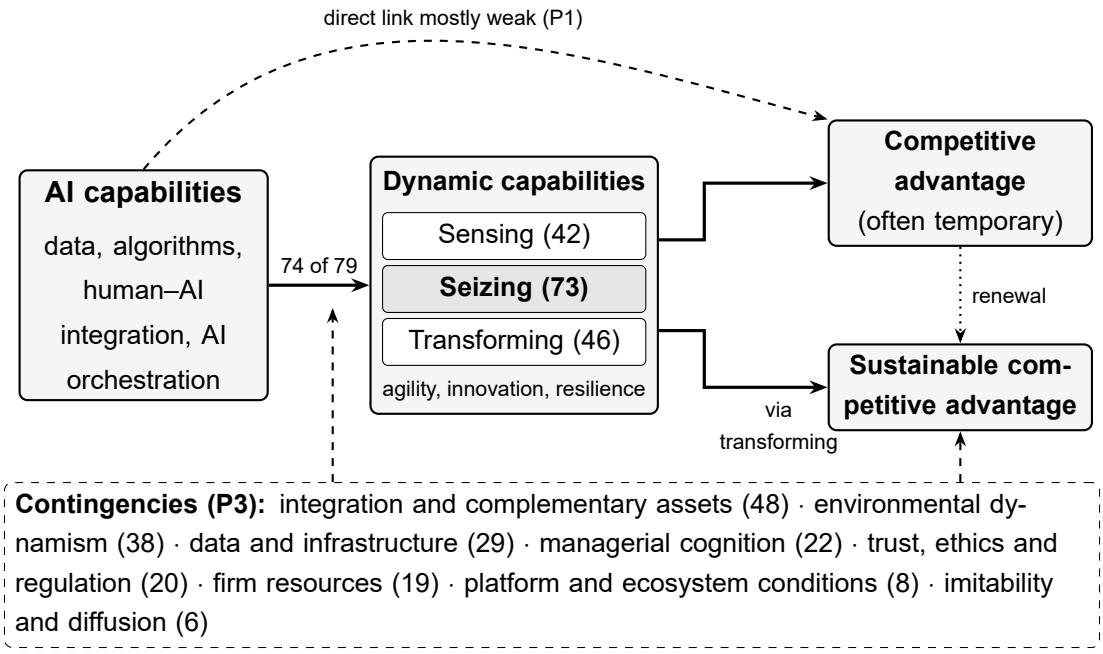


FIGURE 4.2: Final Evidence-Informed Conceptual Model

Source: Author's own work based on the conceptual framework (Figure 2.3) and the synthesis of 109 studies; numbers indicate supporting studies

First, the findings confirm the resource-based argument that AI technologies as such are unlikely to provide a sustained advantage. According to Barney (1991, p. 99), a resource must be valuable, rare, difficult to imitate and non-substitutable to generate sustained competitive advantage. The reviewed evidence suggests that AI tools increasingly fail the rarity and imitability criteria, because generative AI is democratised (Cui et al., 2024, p. 583) and freemium tools are accessible even to small retailers (Bernardo, 2026, p. 76). The non-significant direct effects in several studies point in the same direction. What appears to be valuable and difficult to imitate is not AI itself, but the firm-specific way in which AI is situated in organisational activities (Kemp, 2024, p. 618) and combined with human capabilities (Krakowski et al., 2023, p. 1425). This supports the argument of Teece (2007, p. 1319) that sustainable advantage rests on the capabilities that orchestrate and renew resources, rather than on the resources alone.

Second, the review reveals an imbalance within the dynamic capabilities mechanism. Seizing, mostly in the form of agility and decision making, dominates the evidence, whereas transforming and sensing are addressed much less often. This imbalance matters because, as Eisenhardt & Martin (2000, p. 1105) argue, dynamic capabilities often show commonalities across firms; if many firms use AI to improve the same seizing routines, the resulting gains are likely to converge towards competitive parity. The "efficiency trap" identified in the panel data (Huang & Yang, 2026, p. 1) illustrates this

risk: AI knowledge more readily supports efficiency than new value propositions, and reconfiguration is required to escape this trap. For sustainable competitive advantage, transforming capabilities such as business model innovation (Sjödin et al., 2021, p. 574; Le et al., 2026, p. 1) and strategic renewal (Bughin, 2026, p. 1242) therefore appear more decisive than the more frequently studied seizing capabilities.

Third, the findings qualify the platform-specific argument developed in Section 2.5. Data network effects can make AI-enabled platforms more valuable as they learn from users (Gregory et al., 2021, p. 534), but algorithm envelopment shows that such advantages can be overcome by competitors with superior algorithms (Chen et al., 2026, p. 187). The core e-commerce studies mainly show short-term benefits, such as efficiency (Malhotra & Kharub, 2025, p. 290; Bernardo, 2026, p. 76), service innovation (Alnahdi, 2026, p. 1) and financial performance (Sivathanu & Pillai, 2026, p. 93). None of them examines whether these benefits persist. Combined with the evidence on contingencies, this suggests that sustainable competitive advantage in e-commerce depends less on the AI capability itself than on the continuous renewal of data, algorithms and customer relationships, which is consistent with the dynamic view of advantage proposed by Teece et al. (1997, p. 509).

Fourth, the review highlights that the conditions surrounding AI are as important as AI itself. Integration with processes and complementary assets is the most frequently reported contingency, followed by environmental dynamism. Interestingly, dynamism does not have a uniform effect: it strengthens the effect of AI capabilities on resource fluidity (Wang et al., 2025, p. 3003), but weakens the mediating role of service innovation (Li et al., 2025, p. 10) and the effects of business model transformation (Le et al., 2026, p. 1). In addition, trust, ethics and regulation emerge as a distinct group of factors. AI washing (Elhajjar & Itani, 2025, p. 519), algorithmic bias, privacy and explainability concerns (Norfleet, 2026, p. 1) and the need for AI governance (Htet et al., 2026, p. 1; Chotia et al., 2026, p. 2227) indicate that an AI-based advantage can be undermined if customers and stakeholders lose trust, a factor that is particularly relevant for data-dependent e-commerce models.

The literature also contains two distinct perspectives on AI and competitive advantage, which have to be contrasted. The dominant, largely optimistic perspective is represented by the many survey studies that report positive associations between AI capabilities and performance or advantage. A smaller, more critical perspective is found mainly in conceptual work and in studies published in leading strategy and management journals. It emphasises that AI shifts rather than simply strengthens the sources of advantage (Krakowski et al., 2023, p. 1425), that AI must be situated in firm-specific activities to create value (Kemp, 2024, p. 618), that democratised AI provides generic

rather than unique insights (Cui et al., 2024, p. 583), that the performance effects of AI in strategy will depend on competitive dynamics (Csaszar et al., 2024, p. 322), and that firms may even misrepresent their AI capabilities (Elhajjar & Itani, 2025, p. 519). The two perspectives are not necessarily contradictory. Survey studies capture the benefits that individual firms realise relative to their own past or to less advanced competitors, whereas the critical studies ask what happens when AI capabilities diffuse across an industry. For the question of sustainable competitive advantage, the critical perspective is therefore particularly relevant, because sustainability concerns precisely the situation in which competitors catch up.

The findings have to be interpreted against the characteristics of the evidence base. Most studies are cross-sectional surveys based on managers' self-assessments, which can establish associations but not causality and may be affected by common method bias. Longitudinal designs are rare, and the geographical concentration in China, the Middle East and South Asia limits the transferability of the results to European, and particularly German, e-commerce markets. Moreover, many studies were published very recently and partly in journals with less established review processes. To address this, the conclusions above were checked against the high-quality studies: the pattern of results is the same when only studies that meet all six quality criteria Q1 to Q6 are considered, which increases confidence in the main findings. Finally, for 72 of the 109 studies the data extraction relies on the abstracts, so that details reported only in the full texts may have been missed. The full-text check of the 37 accessible studies shows that this risk is real: it changed the coding of one study from an unaddressed to a non-significant direct effect (Wang et al., 2026, p. 1), revealed a sample size that differs from the abstract (Sukati et al., 2026, p. 1), showed that the results of one study are based on simulated values (Progga & Ali, 2026, p. 1) and raised the quality score of two studies whose analyses are only described in the full text (Dubey et al., 2020, p. 1; Csaszar et al., 2024, p. 322).

Relating the results to the research gaps identified in Table 2.3 shows what the review contributes and what remains open. Regarding G1, the review confirms that e-commerce and platform evidence is scarce (18 of 109 studies) and adds a structured synthesis of what this evidence shows. Regarding G2, the separate coding of outcome types demonstrates that most studies measure performance or competitive advantage, while sustainable competitive advantage is rarely examined and never in the core studies. Regarding G3, the review shows that sensing, seizing and transforming are addressed very unevenly and that transforming appears most relevant for sustainability. Regarding G4, it identifies eight families of contingencies, including platform conditions and trust, which have so far been studied in isolation. Finally, regarding G5, the review

provides the integrated synthesis of AI capabilities, dynamic capabilities and sustainable competitive advantage that was missing in earlier reviews. The gaps concerning the long-term persistence of advantage and the European context, however, cannot be closed with the existing evidence and are therefore taken up as avenues for future research in Chapter 5.

For e-commerce and platform-based businesses, the findings have several implications. First, firms should not expect that the adoption of AI tools alone will differentiate them, since such tools are increasingly accessible to competitors. Second, the evidence points to a sequence in which AI capabilities are first converted into agility and service or marketing innovation and then, through business model innovation and reconfiguration, into an advantage that can be renewed. E-commerce firms that use AI mainly for efficiency, for example in logistics or customer service, may therefore achieve performance gains that competitors can quickly imitate, whereas firms that use AI to develop new value propositions and business models are more likely to sustain an advantage. Third, complementary assets matter: digital infrastructure (Alnahdi, 2026, p. 1), data quality (Kokash, 2026, p. 1), integration with suppliers and customers (Zhong, 2026, p. 1; Malhotra & Kharub, 2025, p. 290) and an innovation-oriented culture (Chen et al., 2022, p. 1) determine whether AI capabilities translate into results. Finally, trust and governance are part of the competitive equation, since customer-facing AI applications in e-commerce depend on the willingness of customers to share data. These implications are developed into managerial recommendations in Chapter 5.

Taken together, the results suggest that the question is not whether firms should invest in AI, since the evidence consistently links AI capabilities to performance and advantage, but how they can turn these capabilities into an advantage that lasts. The evidence-informed model in Figure 4.2 provides an answer: through dynamic capabilities, especially transforming capabilities, and under organisational and contextual conditions that allow the advantage to be continuously renewed. Chapter 5 draws on these findings to answer the research questions and to derive theoretical contributions and managerial recommendations.

5 Conclusion and Recommendations

This final chapter brings the thesis to a close. It first summarises the study and answers the three research questions (Section 5.1). It then outlines the theoretical contributions (Section 5.2) and derives recommendations for managers of e-commerce and platform-based businesses (Section 5.3). Section 5.4 reflects critically on the limitations of the thesis and identifies avenues for future research, before Section 5.5 concludes.

5.1 Summary and Answers to Research Questions

The starting point of this thesis was a tension observed in Chapter 1. On the one hand, AI is widely regarded as a powerful source of value in data-rich digital businesses. On the other hand, AI technologies are becoming broadly accessible, and many firms struggle to turn their AI investments into measurable returns. The aim of the thesis was therefore to examine, on the basis of the existing academic literature, how AI capabilities contribute to sustainable competitive advantage in e-commerce and platform-based digital businesses, and which role dynamic capabilities play in this process.

To achieve this aim, the thesis combined the resource-based view and dynamic capabilities theory into a conceptual framework (Chapter 2) and conducted a systematic literature review (Chapter 3). Two searches in the EBSCO Discovery Service identified 612 records, of which 109 studies were included after de-duplication, screening and eligibility assessment. Eighteen of these are core studies from e-commerce, online retail and platform contexts, and 91 are supporting studies from other industries. The evidence was extracted with a structured framework, checked through an independent second coding and, where accessible, against the full texts, assessed for quality and synthesised thematically along the research questions (Chapter 4). Table 5.1 summarises the answers to the three research questions.

TABLE 5.1: Summary of Answers to the Research Questions

Research question	Answer	Section
RQ1: How are AI capabilities associated with competitive advantage?	Positively, in the large majority of studies and in the core e-commerce studies, but mostly indirectly; direct effects often become weak once mediators are considered. Evidence concerns mainly performance and competitive advantage, rarely sustainable advantage.	4.2, 4.5
RQ2: How do sensing, seizing and transforming relate to this relationship?	Dynamic capabilities are the main mechanism through which AI capabilities create value. Seizing (agility, innovation) dominates the evidence, transforming and sensing are studied much less often, although transforming appears most relevant for sustainability.	4.3, 4.4
RQ3: Which factors enable or constrain sustainable advantage through AI?	Integration with processes and complementary assets, environmental dynamism, data and infrastructure, managerial cognition, trust, ethics and regulation, firm resources, platform conditions and imitability. Their effect on the persistence of advantage over time is not yet established.	4.5, 4.6

Source: Author's own work

With regard to *RQ1*, the review shows that AI capabilities are associated with competitive advantage and firm performance in a consistent way. Sixty-three of the 79 studies that examine this link report a positive association, and the same holds for 12 of the 14 core studies from e-commerce and platform contexts, while the other two describe the link as conditional. However, the association is predominantly indirect. In several studies, the direct effect of AI capabilities is not significant once mediating capabilities are taken into account (Alioune, 2026, p. 1; Akaelu et al., 2026, p. 1; Wang & Zhang, 2026, p. 1; Kwiatek, 2025, p. 1; Wang et al., 2026, p. 1). Furthermore, most studies measure firm performance or competitive advantage at one point in time. Firm performance and competitive advantage were coded separately from sustainable competitive advantage, and only 11 studies examine the latter; none of them is set in e-commerce. The answer to RQ1 is therefore that AI capabilities are associated with competitive advantage, but that the existing evidence does not yet demonstrate that this advantage is sustainable.

With regard to *RQ2*, the review provides strong support for the dynamic capabilities perspective. Seventy-four of the 79 studies that address the link report that AI capabilities strengthen sensing, seizing or transforming, or that these capabilities explain how AI creates value. The mechanism is, however, uneven. Seizing, in particular in the form of organisational, strategic and supply chain agility and of service and marketing innovation, is addressed most often. Transforming, in the form of business model innovation and reconfiguration, is addressed less often, but the evidence suggests that it

is decisive for moving beyond efficiency gains towards new value propositions (Huang & Yang, 2026, p. 1). Sensing is addressed about as often as transforming and far less often than seizing, although AI appears naturally suited to support it. The answer to RQ2 is therefore that dynamic capabilities, and in particular transforming capabilities, are the mechanism through which AI capabilities can be converted into a sustainable advantage.

With regard to *RQ3*, the review identifies eight families of factors that enable or constrain the creation of advantage through AI capabilities. The most frequently reported are the integration of AI into business processes and the availability of complementary assets, followed by environmental dynamism, data quality and digital infrastructure, managerial cognition, and trust, ethics and regulation. Firm resources and characteristics, platform and ecosystem conditions and the imitability of AI applications complete the picture. These factors explain why identical AI investments lead to different results in different firms. Because the evidence base contains few longitudinal studies, however, it remains open which of these factors determine whether an AI-based advantage persists over time.

Taken together, the central conclusion of the thesis is that AI capabilities are an important, but not a sufficient, condition for sustainable competitive advantage. AI capabilities create value when firms convert them into dynamic capabilities, especially into the ability to transform their business models and resource base, and when complementary assets, governance and managerial cognition support this conversion. In e-commerce and platform businesses, where AI tools are widely accessible and competitors can quickly imitate efficiency gains, this conversion is likely to be the main source of a lasting advantage.

5.2 Theoretical Contributions

The thesis makes five contributions to research at the intersection of strategic management and information systems.

First, it integrates the resource-based view and dynamic capabilities theory into a coherent explanation of how AI creates competitive advantage. The review shows that AI capabilities rarely meet the criteria of rareness and inimitability that Barney (1991, p. 99) identified as conditions for sustained competitive advantage, because AI tools are becoming democratised (Cui et al., 2024, p. 583). Their value lies instead in the way they are situated in firm-specific activities (Kemp, 2024, p. 618) and in the dynamic capabilities through which they are deployed and renewed. This supports the argument of Teece (2007, p. 1319) that sustainable advantage rests on the capabilities

that orchestrate and reconfigure resources, and it applies this argument specifically to AI.

Second, the thesis contributes to conceptual clarity by distinguishing systematically between firm performance, competitive advantage and sustainable competitive advantage. By coding these outcomes separately, the review reveals that the literature frequently treats performance gains as evidence of competitive advantage, while sustainable advantage is rarely measured. This distinction, which responds to the conceptual ambiguity noted by Wu et al. (2026, p. 1), is a precondition for cumulative research on AI and long-term competitiveness.

Third, the thesis refines the dynamic capabilities mechanism for AI. Rather than treating dynamic capabilities as an aggregate construct, it shows that sensing, seizing and transforming are addressed very unevenly and that their contributions differ. The finding that seizing dominates the evidence, while transforming appears most relevant for sustainability, connects the AI literature with the argument of Eisenhardt & Martin (2000, p. 1105) that dynamic capabilities often show commonalities across firms. If many firms use AI to improve similar seizing routines, the resulting gains are likely to converge, and only the reconfiguration of business models and resource bases can create lasting differences.

Fourth, the thesis provides a structured map of the contingencies that shape the effect of AI capabilities. Previous studies have examined individual moderators in isolation. The review integrates them into eight families and adds two groups that are particularly relevant for digital business but have received little attention in the capability literature: trust, ethics and regulation, and platform and ecosystem conditions such as data network effects and algorithm competition (Gregory et al., 2021, p. 534; Chen et al., 2026, p. 187).

Fifth, the evidence-informed conceptual model in Figure 4.2 offers a basis for future empirical research. It specifies the expected pathways from AI capabilities through the three dynamic capabilities to competitive and sustainable competitive advantage, and it identifies the boundary conditions that should be considered. In addition, the distinction between core and supporting evidence, combined with a transparent quality assessment, provides a methodological template for reviews in fields where context-specific evidence is still scarce.

5.3 Managerial Recommendations

The findings lead to six recommendations for managers of e-commerce and platform-based businesses who want to turn AI into a lasting competitive advantage. They are

summarised in Table 5.2 and explained below.

TABLE 5.2: Managerial Recommendations

Recommendation	Rationale	Supporting evidence
1. Treat AI as an organisational capability, not as a tool	AI tools are accessible to competitors; value lies in firm-specific combinations	(Cui et al., 2024, p. 583; Kemp, 2024, p. 618; Mikalef & Gupta, 2021, p. 1)
2. Invest in complementary assets	Data quality, infrastructure and integration determine the effect of AI	(Kokash, 2026, p. 1; Alnahdi, 2026, p. 1; Zhong, 2026, p. 1)
3. Use AI to transform, not only to optimise	Efficiency gains are easily imitated; new value propositions last longer	(Huang & Yang, 2026, p. 1; Sjödin et al., 2021, p. 574; Le et al., 2026, p. 1)
4. Build agility and resilience	Agility and resilience are the main channels from AI to performance	(Fosso Wamba, 2022, p. 1; Wang & Zhang, 2026, p. 1; Elshaer et al., 2025, p. 1)
5. Make trust and governance part of the AI strategy	Customer data and trust underpin AI-based advantage in e-commerce	(Elhajjar & Itani, 2025, p. 519; Norfleet, 2026, p. 1; Chotia et al., 2026, p. 2227)
6. Develop managerial AI cognition and monitor competition	Managers' framing of AI and competitors' algorithms shape the persistence of advantage	(Wu et al., 2026, p. 1; Chen et al., 2026, p. 187; Csaszar et al., 2024, p. 322)

Source: Author's own work based on the findings in Chapter 4

Treat AI as an organisational capability. Managers should avoid equating the purchase or adoption of AI tools with an AI strategy. Since many AI applications, including generative AI, are available to all competitors, they are unlikely to create a lasting difference on their own. Instead, managers should build an AI capability in which several AI-specific resources are combined in a firm-specific way (Mikalef & Gupta, 2021, p. 1) and situate AI in the specific activities and decisions of their business (Kemp, 2024, p. 618).

Invest in complementary assets. The effect of AI capabilities depends on the quality of data (Kokash, 2026, p. 1), on digital infrastructure (Alnahdi, 2026, p. 1) and on the

integration of AI with suppliers and customers (Zhong, 2026, p. 1). For e-commerce firms, this means that investments in AI should be accompanied by investments in data governance, system integration and logistics, which are often less visible but decisive for success.

Use AI to transform, not only to optimise. Many e-commerce firms use AI mainly to increase efficiency, for example in logistics, pricing or customer service. Such gains are valuable but can be imitated quickly. The evidence suggests that firms should also use AI to develop new value propositions and business models, for example AI-based services, personalised offerings or platform-based ecosystems, because reconfiguration is required to escape the efficiency trap (Huang & Yang, 2026, p. 1) and to scale AI (Sjödin et al., 2021, p. 574).

Build agility and resilience. Agility and resilience are among the most frequent channels through which AI capabilities improve performance and competitive advantage (Fosso Wamba, 2022, p. 1; Wang & Zhang, 2026, p. 1). Managers should therefore use AI-generated insights to accelerate decision making and to adapt supply chains, assortments and marketing quickly to changing conditions, and they should design organisational structures and routines that allow such adaptation.

Make trust and governance part of the AI strategy. Customer-facing AI in e-commerce depends on customers' willingness to share data. Managers should therefore communicate transparently about their use of AI, address privacy, bias and explainability concerns (Norfleet, 2026, p. 1), implement ethical AI governance (Chotia et al., 2026, p. 2227) and avoid overstating their AI capabilities, since AI washing can damage trust and reputation (Elhajjar & Itani, 2025, p. 519).

Develop managerial AI cognition and monitor competition. The way in which managers interpret AI, as a tool, an actor or a strategic resource, shapes whether a firm achieves fleeting efficiency gains or sustained advantages (Wu et al., 2026, p. 1). Managers should therefore invest in their own AI literacy and in that of their teams. At the same time, they should monitor competitors' AI capabilities closely, since an entrant with a superior algorithm can overcome even a large user base (Chen et al., 2026, p. 187), and the performance effects of AI depend on how quickly competitors adopt similar capabilities (Csaszar et al., 2024, p. 322).

The six recommendations are not independent measures but can be understood as a sequence. In the short term, firms should establish the foundations of an AI capability, namely reliable data, digital infrastructure and the integration of AI into core processes, and they should define governance rules for the responsible use of customer data. In the medium term, they should use these foundations to build agility, for example

by accelerating decisions on assortments, prices and marketing campaigns, and by using AI to develop new services and customer experiences. In the long term, they should use the resulting insights and capabilities to transform their business models and ecosystem relationships, since this is where the evidence locates the most durable sources of advantage. Throughout this sequence, managers should regularly assess whether their AI-based benefits are still distinctive or whether competitors have caught up, because this assessment indicates when the next cycle of renewal is required.

5.4 Limitations and Future Research

The findings of this thesis are subject to several limitations, some of which arise from the review method and some from the state of the evidence. Regarding the method, the search was conducted in a single discovery service, restricted to English-language, peer-reviewed journal articles, and the screening was carried out by a single reviewer. Most importantly, the data extraction for 72 of the 109 studies was based on the abstracts, complemented by bibliographic information; only the 37 accessible open-access studies could be checked against their full texts. Abstracts report the main findings but omit many details, for example on measurement, sample characteristics or limitations. Whether limitations are discussed could therefore be assessed for only 42 studies and was reported separately from the quality score instead of being scored as not met, and some findings may be simplified. These limitations were mitigated by explicit criteria, reason codes, a documented screening log and an AI-assisted second coding with adjudication of all disagreements, but they should be addressed in future work through a complete full-text analysis and independent coding by several human reviewers.

Regarding the evidence, the review is limited by the characteristics of the primary studies. Most of them are cross-sectional surveys based on managers' perceptions, which do not allow causal conclusions and may be affected by common method bias. The constructs used to measure AI capabilities, dynamic capabilities and competitive advantage differ considerably, which prevents a quantitative comparison of effect sizes. The evidence is geographically concentrated in China, the Middle East and South Asia, and none of the studies that report their context is set in Germany. Finally, three quarters of the included studies were published in 2025 and 2026, so that the evidence base is likely to change quickly.

Looking back on the research process, several decisions of the author also shaped the results and deserve critical reflection. The two-step search strategy kept the review manageable, but it relied on the term *AI capability* in the primary search, which is why 16 relevant studies had to be added through other methods; a broader string with a

larger review team might have found further studies. The decision not to use journal rankings as an inclusion criterion increased the coverage of the evidence, but it also brought studies of very different quality into the sample, which made the quality assessment more important and the synthesis more complex. The distinction between core and supporting studies proved useful for separating context-specific from general evidence, but the boundary between the two tiers is not always sharp, for example for retail or supply chain studies with an e-commerce component. Finally, the conceptual framework derived in Chapter 2 guided the coding and may have directed attention towards dynamic capabilities at the expense of alternative explanations, such as institutional or behavioural theories, which some of the included studies also use. Being aware of these choices is important for interpreting the findings, and future reviews could address them by combining several search strategies, coders and theoretical lenses.

These limitations point to several avenues for future research:

- *Longitudinal research on sustainability.* Future studies should observe firms over several years to examine whether and how long AI-based advantages persist, using panel data or repeated surveys, and should measure sustainable competitive advantage directly rather than inferring it from performance.
- *E-commerce and platform-specific studies.* Since only 18 of the 109 studies are set in e-commerce or platform contexts and none examines sustainable advantage, research should test the evidence-informed model in these contexts, including the role of data network effects and algorithm competition.
- *European and German evidence.* Studies of European and particularly German e-commerce firms would show whether the findings hold in markets with different regulatory conditions, such as stricter data protection.
- *Differentiated dynamic capabilities.* Future research should examine sensing, seizing and transforming separately and compare their contributions, with particular attention to the under-researched sensing capability and to the transforming capabilities that appear most relevant for sustainability.
- *Generative and agentic AI.* As generative and agentic AI diffuse rapidly, research should examine whether these technologies change the conditions of imitability and the sources of advantage identified in this review.
- *Meta-analysis and full-text reviews.* Once the evidence base matures, a meta-analysis could quantify the strength of the relationships, and a full-text review with several coders could validate the findings of this abstract-based synthesis.

5.5 Final Conclusion

This thesis set out to understand how AI capabilities contribute to sustainable competitive advantage in e-commerce and platform-based digital business. Based on a systematic review of 109 studies, it concludes that AI capabilities are consistently associated with competitive advantage, but that this advantage arises mainly indirectly, through dynamic capabilities such as agility, innovation and the transformation of business models, and depends on complementary assets, governance, managerial cognition and the competitive environment. At the same time, the review shows that the literature has so far focused on performance gains at a single point in time and has rarely examined whether AI-based advantages last, especially in e-commerce.

For research, the thesis offers an evidence-informed model that links AI capabilities, sensing, seizing and transforming, and sustainable competitive advantage, and identifies the conditions under which this link holds. For practice, it suggests that e-commerce and platform firms should not ask whether to use AI, since AI is quickly becoming a standard tool, but how to build the capabilities that allow them to renew their advantage faster than competitors can imitate it. In a market in which algorithms, data and customer expectations change continuously, the most durable advantage may not be a particular AI capability, but the dynamic capability to keep reinventing how AI creates value.

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A Search Strings

The searches were carried out in the EBSCO Discovery Service (IU university library) on 30 October 2026, using the filters *Peer reviewed* and *publication date 2015–2026*. The field codes `TI` (title) and `AB` (abstract) restrict the search to the respective fields.

Search S1 (primary search)

```
(TI ("AI capabilit*" OR "artificial intelligence capabilit*" OR "AI-enabled
capabilit*" OR "AI-driven capabilit*") OR AB ("AI capabilit*" OR "artificial
intelligence capabilit*" OR "AI-enabled capabilit*" OR "AI-driven capabilit*"))
AND (TI ("dynamic capabilit*" OR "competitive advantage" OR "sustainable
competitive advantage") OR AB ("dynamic capabilit*" OR "competitive advantage"
OR "sustainable competitive advantage"))
```

Records retrieved: 407; after automatic de-duplication: 214.

Search S2 (supplementary search, e-commerce and platform context)

```
(TI ("artificial intelligence" OR "AI-enabled" OR "AI-driven" OR "generative
AI") OR AB ("artificial intelligence" OR "AI-enabled" OR "AI-driven" OR
"generative AI")) AND (TI ("dynamic capabilit*" OR "competitive advantage") OR
AB ("dynamic capabilit*" OR "competitive advantage")) AND (TI ("e-commerce" OR
"electronic commerce" OR "online retail*" OR "e-retail*" OR "digital platform*"
OR "platform business*" OR "platform ecosystem*") OR AB ("e-commerce" OR
"electronic commerce" OR "online retail*" OR "e-retail*" OR "digital platform*"
OR "platform business*" OR "platform ecosystem*"))
```

Records retrieved: 205; after automatic de-duplication: 106.

B Overview of the Included Studies

The table lists all 109 studies included in the systematic review, ordered by tier and author. Design: Concept. = conceptual, Qual./case = qualitative or case study, Panel = panel or archival data, Mixed = mixed methods, Model = formal model. Outcome: FP = firm performance, CA = competitive advantage, SCA = sustainable competitive advantage, BMI = business model innovation, Innov. = innovation, Resil. = resilience. Score: quality score on criteria Q1–Q6 of Table 3.4 (0–6). Q7: limitations discussed (yes, no, n.a. = not assessable from the abstract). IDs marked with an asterisk were checked against the full text; all other codes are based on the abstract. VHB: journal listed in the VHB Rating 2024 (descriptive only).

ID	Study	Design	Context	Outcome	Score	Q7	VHB
Core (e-commerce and platforms) studies (n = 18)							
R186	Aljarboa (2024, p. 1)	Survey	E-commerce SMEs	FP	5	yes	no
R111	Alnahdi (2026, p. 1)	Survey	Retail, e-commerce	FP	5	n.a.	no
R242*	Awad (2024, p. 170)	Survey	E-retail	Innov.	5	yes	no
R228*	Awamleh & Bustami (2022, p. 1)	Survey	E-commerce	CA	4	yes	no
R193*	Bernardo (2026, p. 76)	Qual./case	Clothing e-commerce	FP	4	no	no
O08*	Bunea & Corboș (2026, p. 1)	Mixed	Electronic retailing	CA	5	yes	no
R182*	Chen et al. (2022, p. 1)	Survey	E-commerce	FP	6	yes	no
O14	Chen et al. (2026, p. 187)	Model	Platforms	CA	6	n.a.	yes
O15*	Gregory et al. (2021, p. 534)	Concept.	Platforms	Other	6	no	yes
R216*	Guo et al. (2025, p. 1250)	Survey	Cross-border e-commerce	CA	6	yes	yes
R184	Haqqani & Aleem (2026b, p. 1)	Survey	Business, e-commerce, finance	FP	5	n.a.	no
R241	Malhotra & Kharub (2025, p. 290)	Survey	E-commerce logistics	FP	5	n.a.	yes
R155	Mittal (2025, p. 1)	Concept.	E-business marketing	Other	3	n.a.	no
R269*	Modgil et al. (2022, p. 1246)	Qual./case	E-commerce supply chain	Resil.	6	yes	yes
R219	Sivathanu & Pillai (2026, p. 93)	Survey	E-commerce	FP	4	n.a.	yes
R213*	Teece & Pitelis (2026, p. 689)	Concept.	MNEs in platform and AI ecosystems	Other	6	no	yes
R088*	Ratna (2026, p. 649)	Concept.	Platform ecosystems	CA	6	no	no

ID	Study	Design	Context	Outcome	Score	Q7	VHB
R199	Zhou et al. (2025, p. 1348)	Model	E-commerce retail	CA	6	n.a.	yes
Supporting (other industries) studies (n = 91)							
R030*	Khattab et al. (2026, p. 1)	Survey	Manufacturing	CA	5	yes	no
R093*	Alhamaideh et al. (2026, p. 1)	Survey	Energy	Resil.	6	yes	no
R161	Abdallah (2025, p. 1)	Concept.	-	CA	1	n.a.	no
R017	Abou-Foul et al. (2023, p. 1)	Mixed	Manufacturing	Other	6	n.a.	yes
R137	Ahmad (2025, p. 194)	Survey	Family-owned small hotels	Other	6	n.a.	no
R011	Akaelu et al. (2026, p. 1)	Panel	SMEs	FP	5	n.a.	no
R025	Akter et al. (2026, p. 1)	Survey	Entrepreneurial firms	FP	6	yes	no
R156	Al-Mamary (2025, p. 1)	Survey	Entrepreneurial ventures	FP	5	n.a.	no
R069	Alaskar (2026, p. 1)	Survey	Multi-industry	Resil.	6	n.a.	no
R003*	Alioune (2026, p. 1)	Survey	High-growth industries	SCA	5	no	no
O06	Alnofeli et al. (2026, p. 1)	Mixed	Banking	CA	6	n.a.	yes
R096	AlQershi (2026, p. 1)	Panel	Logistics, energy, shipping	Resil.	6	n.a.	no
R079*	Ayoub & Sopuru (2026, p. 1)	Survey	Multi-industry	FP	6	yes	no
R119*	Bhatti & Pardaev (2026, p. 1)	Survey	Financial institutions	FP	5	yes	no
O13	Black et al. (2024, p. 1)	Qual./case	Multi-industry	BMI	5	n.a.	yes
R085	Bughin (2026, p. 1242)	Survey	Multi-industry	Other	2	n.a.	yes
R173	Cao et al. (2025, p. 1)	Mixed	Retail	FP	6	n.a.	yes
O05	Chatterjee et al. (2021, p. 205)	Survey	B2B	CA	2	n.a.	yes
R151*	Chatterjee et al. (2024, p. 967)	Survey	B2C relationship management	Other	3	yes	yes
R043	Chotia et al. (2026, p. 2227)	Survey	AI-intensive start-ups	Other	6	n.a.	yes
R008	Chu & Chan (2026, p. 1)	Concept.	MNEs, global value chains	Other	5	n.a.	no
R129*	Csaszar et al. (2024, p. 322)	Mixed	Start-ups, investors	Other	6	no	yes
O16	Cui et al. (2024, p. 583)	Concept.	Brands	SCA	3	n.a.	yes
O12	Dai & Zhang (2026, p. 1)	Panel	Multi-industry	Resil.	6	n.a.	yes
R170	Deryl et al. (2025, p. 1)	Concept.	Branding	SCA	3	n.a.	yes
R040	Dinh (2026, p. 84)	Mixed	Manufacturing SMEs	Innov.	5	n.a.	yes

ID	Study	Design	Context	Outcome	Score	Q7	VHB
R102*	Dubey et al. (2020, p. 1)	Survey	Manufacturing	FP	6	yes	yes
R135	El Maghawry Ibrahim (2024, p. 61)	Survey	Multi-industry	FP	4	n.a.	no
R148	Elhajjar & Itani (2025, p. 519)	Concept.	Marketing	CA	5	yes	yes
R108*	Elshaer et al. (2025, p. 1)	Survey	Hotels	CA	6	yes	no
O04	Fosso Wamba (2022, p. 1)	Survey	Supply chains	FP	6	yes	yes
R168	Gabelaia & Hendieh (2026, p. 657)	Mixed	SMEs	FP	2	yes	yes
R075	Gafsi (2026, p. 1)	Survey	Family firms	CA	6	n.a.	no
R131*	Gao et al. (2025b, p. 1)	Survey	Manufacturing	Innov.	6	yes	no
R028	Ghosh (2025, p. 1)	Qual./case	Industrial	BMI	4	n.a.	yes
R053*	Giang et al. (2026, p. 1)	Survey	SMEs	CA	6	yes	no
R007	Haqqani & Aleem (2026a, p. 1)	Survey	Finance, manufacturing, high-tech	FP	6	n.a.	no
O10	Hossain et al. (2022, p. 240)	Mixed	Manufacturing (export)	SCA	4	n.a.	yes
R001	Htet et al. (2026, p. 1)	Concept.	Marketing	SCA	6	n.a.	no
O11	Huang & Yang (2026, p. 1)	Panel	Multi-industry	Other	4	n.a.	yes
R114*	Javed et al. (2026, p. 1)	Survey	Manufacturing	FP	6	yes	no
R064*	Kao (2026, p. 109)	Qual./case	Multi-industry	Other	4	yes	yes
R087	Kemp (2024, p. 618)	Concept.	-	CA	5	n.a.	yes
R050	Kokash (2026, p. 1)	Survey	Tourism	FP	5	n.a.	no
O03	Krakowski et al. (2023, p. 1425)	Panel	Chess (controlled setting)	CA	4	n.a.	yes
R247*	Kubiczek et al. (2025, p. 199)	Mixed	Multi-industry	CA	3	yes	no
R081*	Kumar et al. (2025b, p. 1)	Survey	Exporting firms	FP	6	yes	yes
R142	Kumar et al. (2025a, p. 1)	Survey	Multi-industry	FP	5	n.a.	yes
R120	Kwiatek (2025, p. 1)	Survey	B2B	FP	6	n.a.	no
R022	Kyriakopoulos et al. (2025, p. 1)	Survey	Multi-industry (NPD)	Innov.	5	n.a.	yes
R037	Le et al. (2026, p. 1)	Survey	Start-ups	Other	5	n.a.	no
R180	Lee et al. (2024, p. 1)	Qual./case	Multi-industry	FP	4	n.a.	no
R004	Li et al. (2025, p. 10)	Survey	B2B industrial	CA	6	n.a.	yes
R132	LI et al. (2025, p. 1)	Mixed	Multi-industry	Innov.	6	n.a.	yes
R031	Li et al. (2026, p. 1)	Survey	Firms in digital transformation	BMI	6	n.a.	yes
R044	Liang & Sun (2026, p. 1)	Panel	Multi-industry	SCA	5	n.a.	yes
R010	Liu & Xin (2026, p. 1)	Panel	Multi-industry	Innov.	4	n.a.	yes
R103*	Maune (2026, p. 6)	Mixed	Multi-industry	FP	5	yes	no

ID	Study	Design	Context	Outcome	Score	Q7	VHB
R077*	McIlveene et al. (2026, p. 1)	Concept.	Small businesses	SCA	6	yes	yes
O01	Mikalef & Gupta (2021, p. 1)	Survey	Multi-industry	FP	4	n.a.	yes
O02	Mikalef et al. (2021, p. 80)	Qual./case	B2B marketing	Other	5	n.a.	yes
R252*	Mohiuddin Babu et al. (2024, p. 2084)	Qual./case	Textile and apparel	Other	2	no	yes
R212*	Morina & Selmonaj (2026, p. 342)	Qual./case	Industrial	CA	4	yes	no
R157	Nayal et al. (2024, p. 137)	Survey	Manufacturing	Resil.	6	n.a.	yes
R181	NAZ et al. (2022, p. 1)	Survey	Food manufacturing	FP	6	n.a.	yes
R174	Naz et al. (2024, p. 2055)	Survey	Hotels	FP	6	n.a.	no
R133	Norfleet (2026, p. 1)	Survey	Multi-industry	CA	4	n.a.	no
R117	Obeidat (2026, p. 1)	Concept.	-	Innov.	6	n.a.	no
R057	Peretz-Andersson et al. (2024, p. 1)	Qual./case	Manufacturing	Other	4	n.a.	yes
R032	Qiao et al. (2026, p. 7274)	Survey	Construction	Innov.	6	n.a.	no
R048*	Ragmoun & Aloulou (2026, p. 1)	Survey	SMEs	Innov.	6	yes	no
R036	Rahman et al. (2026, p. 1)	Survey	Manufacturing, services	Resil.	6	n.a.	no
R110*	Ruangkanjanases et al. (2025, p. 1)	Survey	Manufacturing	CA	5	yes	yes
R122*	Salem et al. (2026, p. 1)	Survey	Banking	FP	6	yes	no
R068*	Schneider et al. (2026, p. 1)	Qual./case	Multi-industry	Other	5	yes	yes
R149	Sharma (2026, p. 1)	Survey	Marketing	SCA	6	n.a.	no
O09	Sjödin et al. (2021, p. 574)	Qual./case	Manufacturing	BMI	4	n.a.	yes
R041*	Sukati et al. (2026, p. 1)	Survey	MSMEs	CA	6	yes	no
R123	Tao et al. (2026, p. 44)	Survey	Manufacturing	Innov.	6	n.a.	yes
R179	Vaillant et al. (2025, p. 1)	Survey	Solution providers, AI platforms	Other	5	n.a.	yes
R183	Wamba-Taguimdje et al. (2020, p. 1893)	Qual./case	Multi-industry	FP	4	n.a.	yes
R018	Wang & Zhang (2026, p. 1)	Mixed	Start-ups	FP	6	n.a.	yes
R056	Wang et al. (2025, p. 3003)	Mixed	Multi-industry	BMI	6	n.a.	yes
R059*	Wang et al. (2026, p. 1)	Survey	High-tech	SCA	6	yes	no
R026	Wei (2026, p. 1)	Concept.	Multi-domain	SCA	5	n.a.	no
R091*	Wishah et al. (2026, p. 1)	Survey	SMEs	FP	5	yes	no
O07	Wu et al. (2026, p. 1)	Concept.	-	SCA	6	n.a.	yes
R098	Xu et al. (2026, p. 1)	Mixed	Construction	Resil.	6	n.a.	no

ID	Study	Design	Context	Outcome	Score	Q7	VHB
R086*	Progga & Ali (2026, p. 1)	Concept.	Enterprises	Innov.	3	no	no
R006	Zhang (2026, p. 1)	Mixed	Multi-industry	CA	6	n.a.	no
R083	Zhong (2026, p. 1)	Survey	Manufacturing	BMI	6	n.a.	yes

Declaration of Honor

I hereby declare that I have written this thesis independently and without the use of any sources or aids other than those stated. All passages taken verbatim or in essence from published or unpublished works of others have been marked as such. This thesis has not been submitted in the same or a similar form to any other examination authority.

Place, Date: _____

Signed: _____